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Inherited inequality and the distribution of opportunities in the USA, China, India and South Africa

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Abstract
Researchers have sought to quantify the extent of inequality that is inherited from previous generations or factors that are predetermined at birth in multiple ways, including a large body of work on intergenerational mobility and inequality of opportunity. Many of the most frequently used approaches to measuring mobility or inequality of opportunity fit within a general framework that involves, as a first step, an estimation of the extent to which inherited personal characteristics can predict current incomes. We suggest a new approach, within that broad framework, that is sensitive to differences across the entire conditional distributions of relevant population subgroups, rather than just in their means. Sensitivity to differences in higher moments of the conditional distributions allows for a comprehensive assessment of inherited inequality. We apply this approach to household income distributions in China, India, South Africa and the USA, to illustrate how the method performs in different settings. We find that inherited inequality accounts for large shares of total inequality, from 36% in the USA to 59% in China, 62% in India, and 81% in South Africa.

KEYWORDS
China, India, inherited inequality, mobility, opportunity, South Africa, transformation trees, USA

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1 INTRODUCTION

People's educational and professional achievements, incomes and wealth are generally not independent of their background. Various attributes that are determined at or before birth or during childhood—such as sex at birth, race, ethnicity or caste, parental income and other aspects of family background—are powerful predictors of a person's own economic outcomes later in life. Large bodies of work have sought to quantify the extent to which these inherited or predetermined characteristics shape people's life outcomes, and to compare results across societies or over time, including the literatures on intergenerational mobility, inequality of opportunity (IOp) and sibling correlations.

This paper contributes to those literatures in three ways.

First, we note that most of these approaches rely on using observed inherited characteristics (often termed 'circumstances') to *predict* future outcomes—hereafter incomes, for simplicity. We suggest a simple general framework for the measurement of inherited inequality that relies on comparisons of inequality in observed and predicted income distributions, and show that some measures in current use are special cases.

In this general framework, we define the hypothetical situation in which there is no inherited inequality as one in which inherited circumstances are *not* predictive of outcomes later in life—that is, current-generation income (y) is distributed independently from those circumstances ($\boldsymbol{c}$): $F(y|\boldsymbol{c}) = F(y)$, for all $\boldsymbol{c}$. Although this independence condition requires that the full conditional distributions of income be identical across population subgroups that share the same circumstances, most commonly used approaches require only a weaker condition on conditional means: $E(y|\boldsymbol{c}) = E(y)$ for all $\boldsymbol{c}$. This second condition is implied by—but does not imply—the stronger independence condition.

Our second and main contribution is therefore to propose a new approach to estimating inherited inequality that captures the extent to which circumstances predict full conditional distributions—rather than just averages—for different population subgroups, and that does so in a statistically efficient manner. Given the central role of prediction in the general framework, we draw on new data-driven (supervised machine learning) techniques, which have been shown to be more accurate out-of-sample predictors than many standard econometric approaches used historically (see, for example, Mullainathan and Spiess 2017). Specifically, we propose to use *transformation trees*: a variant of regression trees proposed by Hothorn and Zeileis (2021) that generates a data-driven partition of the population into groups with homogeneous inherited characteristics, while also predicting their conditional distribution functions.

The use of transformation trees represents an optimal solution to a key methodological problem: the choice of the level of disaggregation at which the analysis should be conducted. An excessively granular subdivision into groups will generate noisy estimates of conditional distributions, leading to an upward bias. Conversely, a conservative partition, based on an excessively narrow specification of relevant circumstance interactions, will result in a downward-biased estimate of the extent to which pre-existing factors shape opportunities and outcomes available later in life.

Our transformation-tree-based approach resolves this trade-off through an algorithm that is designed to select partitions optimally—in a well-defined statistical sense—given the available data: it trades off the upward and downward biases so as to maximize a weighted sum of log-likelihood functions (see Section 3). In addition, by using Bernstein polynomials to fit parametric estimates of conditional distributions, the method uses data more efficiently, and leads to much finer quantile groupings than was possible in earlier approaches. It is therefore suitable for any empirical exercise where the objective is to identify the heterogeneity of conditional distributions across latent groups. Moreover, the results can also be interpreted in the spirit of alternative inequality decompositions, in which the between-group term is not independent of within-group inequality.[1]

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Our third contribution is to discuss the phenomenon of inherited inequality in four important countries over many years: China (every two years between 2010 and 2018), India (2005 and 2012), South Africa (2008, 2012, 2015 and 2017), and the USA (every two years between 1970 and 2018). These four countries include the world's two largest nations by population (India and China), as well as the world's two largest economies by GDP (USA and China). South Africa is a significant developing country and arguably the world's most unequal nation. These are four large economies characterized by very different social structures and territorial features.

Using biological sex, parental education, parental occupation, place of birth and ethnicity/race/caste as circumstances, we construct transformation trees to estimate the full conditional distributions of circumstance–homogeneous population subgroups (known as *types* in the IOp literature) for each country, and to compute summary measures of inherited inequality. We find that a substantial amount of inequality is inherited in all four countries, ranging from 14 Gini points (or 36% relative to total inequality) in the USA to 50 Gini points (81%) in South Africa. China and India display intermediate levels of inherited inequality, but the structure of the distribution of opportunities differs: China shows a more pronounced concentration of opportunity at the very top of the distribution, while India has a large share of the population with extremely limited access to opportunity at the bottom.

Transformation trees can also be used to assess the relative importance of different circumstances in shaping the income distribution. By aggregating (bagging) hundreds of trees, we derive Shapley values that quantify the average relative importance of each ascriptive characteristic. Our results indicate that all circumstances play a meaningful role, with race and caste being especially influential in South Africa and India, respectively. Moreover, the 'marginal effects' of each category within each circumstance are obtained from the predicted conditional distributions. These partial effects find high premia associated with being White in South Africa, being born in specific regions of China (e.g. Shanghai and Zhejiang), and reporting higher parental education in India. In the USA, the most negative marginal effect is associated with identifying as Black.

The paper proceeds as follows. Section 2 briefly describes a general framework for the estimation of inherited inequality, of which some of the most common approaches in the measurement of mobility and IOp are shown to be special cases. Section 3 discusses the key model selection challenge faced by these methods, and introduces our own approach to estimating inherited inequality as another special case within the same general framework, albeit one designed specifically to optimally address the model selection issue.

Section 4 describes the data, and Section 5 presents results. These results include not only summary estimates of inherited inequality in the four selected countries, but also several complementary statistical and visualization tools to help the reader to understand the complexity of the phenomenon: (i) a schematic description of the population partition that reveals the most salient cleavages in each society (again, in a well-defined statistical sense); (ii) estimates of the conditional cumulative distribution functions (CDFs) by type; (iii) a Shapley–Shorrocks decomposition of the average predictive importance of individual circumstances in the overall decomposition; and (iv) a calculation of the marginal influence of each individual characteristic in predicting opportunities. This rich set of by-products of the headline estimates is another advantage of our proposed approach: taken together, this set of statistical tools enables a deeper understanding of inherited inequality based on survey data. Section 6 concludes.

2 | INHERITED INEQUALITY: A SIMPLE GENERAL FRAMEWORK

Consider a population of N individuals, indexed by $i \in \mathcal{N} = \{1, \ldots, N\}$, each of whom is characterized by a current-generation outcome y_i, $y \in \mathbb{R}$, and a set of inherited characteristics, which we call circumstances (following Roemer 1998). For individual i, these circumstances are represented by a k-dimensional vector $\boldsymbol{c_i}$. Let $\boldsymbol{y}$ denote the N-dimensional outcome (or income) vector with entries y_i, $i \in \mathcal{N}$, and let $\boldsymbol{C}$ denote the $N \times k$ matrix with rows $\boldsymbol{c_i}$.

In general, many people may share the same vector of circumstances, so many of the rows of the matrix $\boldsymbol{C}$ may be identical. Without loss of generality, let the number of *distinct* rows of $\boldsymbol{C}$ be denoted by M, for $M \leq N$. If a 'type' is defined as a group of individuals who share identical circumstances, then this means that there are M types. The population can then be exhaustively partitioned into a set of types $T = \{\tau_1, \dots, \tau_m, \dots, \tau_M\}$, where $\tau_m := \{i | \boldsymbol{c_i} = \boldsymbol{c_m}\}$. Let $C = \{c_1, \dots, c_m, \dots, c_M\}$ denote the corresponding set of distinct circumstance vectors, and let $\boldsymbol{c}$ denote the generic random vector in C. Let $T \in \mathbb{T}$, which denotes the set of all possible partitions of the set $\mathcal{N}$, and $C \in \mathbb{C}$, the corresponding set of all possible type circumstance vectors. Note that an exhaustive partition implies that $\bigcup_1^M \tau_m = \mathcal{N}$ and $\bigcap_1^M \tau_m = \emptyset$.

That is the basic setup. Let us now define the benchmark situation in which there is no inherited inequality as one in which $\boldsymbol{y}$ and $\boldsymbol{C}$ are stochastically independent, in the sense that there are no differences across the conditional income distributions of all types:

$$F(y|\boldsymbol{c_l}) = F(y|\boldsymbol{c_m}), \quad \forall \boldsymbol{c_l}, \boldsymbol{c_m} \in C. \tag{1}$$

Given full stochastic independence—that is, when equation (1) holds—C has no predictive power over y. Conversely, if equation (1) does not hold, then the associations between $\boldsymbol{C}$ and $\boldsymbol{y}$ across the population imply that circumstances C have (some) predictive power over y. That is, there exist non-constant prediction functions

$$y = f(\boldsymbol{c}, \varepsilon), \quad f \in \mathcal{F}, \tag{2}$$

that outperform constant functions in predicting y out of sample. In equation (2), ε denotes a random variable that captures other influences on y and is the residual term in the prediction model, and $\mathcal{F} : \mathbb{C} \to \mathbb{R}$ denotes the set of possible prediction functions linking circumstances to outcomes.

Since the benchmark situation of zero inherited inequality is characterized by C having no predictive power over y, it is natural to think of inherited inequality as the extent to which circumstances do, in fact, predict the outcome y in a particular society. In other words, given a prediction function $f \in \mathcal{F}$, absolute inherited inequality can be defined simply as $I_n^A(y, \boldsymbol{c}, f) = I(\widehat{y})$, where $\widehat{y} = \widehat{f}(\boldsymbol{c})$. Absolute inherited inequality is simply inequality in the distribution of predicted incomes, when incomes are predicted by circumstances. One can also define relative measures of inherited inequality as $I_n^R(y, \boldsymbol{c}, f) = I(\widehat{y})/I(y)$, where $\widehat{y} = \widehat{f}(\boldsymbol{c})$. Relative inherited inequality is the ratio—or potentially a monotonically increasing function of the ratio—of inequality in predicted incomes to inequality in observed incomes.

Indeed, it turns out that most methods for estimating the intergenerational transmission of advantage currently in use—including relative measures of intergenerational mobility and inequality of opportunity—revolve around estimating prediction models of the general form (2), using different functions in the set of possible functions $\mathcal{F}$, then computing objects analogous to $I_n^A(y, \boldsymbol{c}, f)$ or $I_n^R(y, \boldsymbol{c}, f)$, often using different inequality indices $I(\cdot)$.[2] Intergenerational mobility and IOp are in fact two special cases of inherited inequality. The former is the workhorse of labour economists studying the intergenerational transmission of earnings and education, while the latter is primarily used by normative economists to analyse unfair inequalities, and is the focus of our empirical analysis in Section 4.

2.1 | Special cases: intergenerational mobility

Suppose, for example, that the only inherited characteristic that really matters is parental income y_p. Then the vector of circumstances reduces to a scalar: $\boldsymbol{c} = y_p$. If, in addition, we choose

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a prediction function $f(\boldsymbol{c}) = f\left(y_p\right)$ of the form $y = \mathrm{e}^{\alpha+\beta \log y_p+\varepsilon}$, which can be estimated through the standard Galtonian regression $\log y = \alpha + \beta \log y_p + \varepsilon$, then we are clearly in the world of intergenerational mobility measurement. See, for example, Solon (1992) and Chetty *et al.* (2014) for classic references.

In that standard formulation, predicted incomes are given by $\hat{y} = \mathrm{e}^{\hat{\alpha}+\hat{\beta}\log y_p}$. Although $\hat{\beta}$, an estimate of the intergenerational elasticity of income, is a common measure of mobility, another frequently used measure is the correlation coefficient between $\log y$ and $\log y_p$, which can be written as $\hat{\beta}\sqrt{\mathrm{var}\log y_p/\mathrm{var}\log y}$. This is, of course, a monotonic function of $I(\hat{y})/I(y)$ when the inequality index is the variance of logarithms.

2.2 | Special cases: IOp

Similarly, absolute and relative estimates of IOp can also be written as examples of $I_n^A(y, \boldsymbol{c}, f)$ or $I_n^R(y, \boldsymbol{c}, f)$, but typically with circumstance vectors with $k > 1$. Inspired by Roemer (1998), Checchi and Peragine (2010) propose to estimate IOp by aggregating income differences *across* the quantiles of the conditional distributions, while abstracting from income differences *within* type-specific distributions. Denoting the overall mean income $E(y)$ by μ, and the mean income for a given quantile q across types by μ_q,[3] their prediction function is given by

$$\hat{y}_{EP} = \hat{f}_{EP}(\boldsymbol{c}) = \frac{\mu}{\mu_q} F^{-1}(q|\boldsymbol{c}). \tag{3}$$

Denote the income at quantile q of the conditional distribution on circumstances $\boldsymbol{c}$, $F^{-1}(q|\boldsymbol{c})$, by y_{qc}. Then their absolute IOp measure $I\left(\hat{y}_{EP}\right) = I\left(\mu y_{qc}/\mu_q\right)$ is simply $I_n^A(y, \boldsymbol{c}, f_{EP})$. Similarly, the relative version is $I_n^R(y, \boldsymbol{c}, f_{EP}) = I\left(\hat{y}_{EP}\right)/I(y)$. In other words, IOp is computed as inequality in predicted incomes after normalizing each observation's income y_{qc} by the average income (across types) of individuals located in the same quantile in their respective conditional distributions (μ_q). Inequality is then calculated across the resulting ratios.[4]

Because it computes IOp as gaps across quantiles of conditional distributions, this approach takes the benchmark situation $F(y|\boldsymbol{c}) = F(y), \forall \boldsymbol{c}$, as corresponding to *equality* of opportunity—much as we defined the absence of inherited inequality in Section 1. In the IOp literature, the use of this benchmark is associated with what is known as *ex post* IOp, whereas measures that compute deviations from the weaker benchmark $E(y|\boldsymbol{c}) = E(y), \forall \boldsymbol{c}$, are known as *ex ante* IOp measures. While we do not dwell on that distinction and do not use that terminology further in this paper, our approach below can be seen as a methodological advance on the Checchi and Peragine (2010) method. We use the more general term 'inherited inequality', but our application belongs to the IOp branch of the literature.[5]

3 | ESTIMATING INHERITED INEQUALITIES USING TRANSFORMATION TREES

3.1 | The model selection problem

Empirical applications attempting to quantify inherited inequality in practice can suffer from a variety of challenges, including data availability, measurement error (particularly in variables such as parental income or occupation) and small sample sizes. More fundamentally, though, they suffer from a model selection problem in the presence of two competing biases. This is particularly true when the researcher does not focus on a single circumstance (e.g. parental

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income), but instead adopts an agnostic approach and attempts to include a comprehensive set of circumstances in the analysis.

The first bias arises from the partial observability of circumstances. It is rather common for data sources that contain information about individual outcomes to also contain various variables describing inherited or predetermined circumstances such as sex, race and socioeconomic background. But the available information almost certainly covers only a strict subset of the set of all relevant inherited factors. Omission of the unobserved circumstances, or indeed of interactions between categories of the variables that one does observe, tends to bias estimates of inherited inequality downwards. See Ferreira and Gignoux (2011) and Roemer and Trannoy (2016) for a discussion in the case of IOp estimation.[6]

On the other hand, a second source of bias arises from the classic problem of overfitting. For a given sample size, saturating the model with many independent variables and their interactions resulting in a detailed description of types will eventually lead to an upward bias in measures of goodness of fit. This overfitting introduces noise into the predictions, inflating the estimated explained variance and thereby biasing upwards the measured variation attributed to circumstances. (See Chakravarty and Eichhorn (1994) for a discussion of the general problem of measuring inequality when the variable of interest contains an error, Bloise *et al.* (2021) for an application to intergenerational mobility estimation, and Brunori *et al.* (2023) for the case of IOp.)

Obtaining a meaningful estimate of $I(\hat{y})/I(y)$ therefore depends crucially on selecting the 'right' model for the prediction function $y = f(\boldsymbol{c}, \varepsilon)$. Of course, what the 'right' model is depends on the nature and purpose of the exercise. If one is estimating a structural model, then guidance from the theory being tested is indispensable, and econometric methods suitable for the estimation of structural parameters should be used. However, when the model is used for prediction, as is the case here, it may very well be that machine-learning methods from data science perform better. See Mullainathan and Spiess (2017) for an excellent discussion of the role of machine learning in economics, and its advantages in prediction problems.

Indeed, machine-learning methods have recently been adopted in the estimation of mobility (Blundell and Risa 2019) and IOp (Brunori *et al.* 2023). However, since such approaches have focused on differences between means, they are not well suited to assessing deviations from the stricter criterion of equal CDFs (equation (1)), whether one interprets such equality as *ex post* equality of opportunity or simply as the absence of inherited inequality. An alternative data-driven approach is needed, and in what follows, we propose the use of one such approach, namely transformation trees.

3.2 | Transformation trees and inherited inequality

As noted in Section 2, our approach to inherited inequality consists of measuring the dispersion across the types' conditional distribution functions at each quantile, and then appropriately aggregating across quantiles. The key ingredient for the approach, therefore, is to estimate the income level at quantile q in type τ_m, that is, the conditional quantile function $y_{qc} = F^{-1}(q|\mathbf{c} = \mathbf{c_m})$, for all m. When data on the joint distribution $\{\mathbf{y}, \mathbf{C}\}$ is not observed for the full population, estimating these conditional quantile—or their inverse, distribution—functions from a sample notionally involves two steps.

First, an optimal type partition $C \in \mathbb{C}$ needs to be selected, trading off the downward bias that arises from combining subtypes into types against the upward bias from overfitting that arises from an excessively fine partition. Second, given a partition C, the conditional quantile functions must be estimated, either parametrically or non-parametrically. Once that has been done, the resulting estimates $\{\tilde{y}_{qc}\}$ can be used to compute quantile-specific inequality levels (across types), which are then suitably aggregated across quantiles.

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Previous attempts to compute inherited inequalities in this way (see Section 2) have typically suffered from two shortcomings. First, the partition $C \in \mathbb{C}$ was chosen arbitrarily. Second, quantiles were computed at a highly aggregated level (e.g. quartiles or deciles) to ensure that there were sufficient observations in each quantile group for a meaningful computation of inequality across types to take place. Indeed, the fact that this approach requires information on the entire conditional distribution $F\left(y_{qc}|c\right)$, rather than merely the mean μ_c of that distribution for each type, makes it more data-intensive, and is one of the reasons why its adoption in empirical applications has been limited so far.

These combined requirements—to choose an optimal type partition given the available dataset, and to estimate conditional distribution functions for each of those types in a data-scarce environment—make this problem well suited to a new variety of tree-based estimator, developed by Hothorn and Zeileis (2021). This estimator, known as a transformation tree, was specifically designed to estimate conditional distributions for terminal nodes of trees. A transformation tree relies on the assumption that there exist 'good enough' parametric approximations to $F\left(y_{qc}|c\right)$. In the limit, they assume that there exist parameters $\theta \in \Theta$ such that.

$$F\left(y_{qc}|c\right) \approx F\left(\tilde{y}_{qc}, \theta(c)\right), \quad \theta : \mathbb{C} \to \Theta. \tag{4}$$

Here, $\theta(c)$ is known as the conditional parameter function, which maps from the space of all possible circumstance vectors on to the space of possible distributional parameters. Under this assumption, the problem of estimating conditional distribution functions for types in the optimal partition, and hence $\left\{\tilde{y}_{qc}\right\}$, reduces to the problem of selecting the optimal parameter estimates $\hat{\theta}$ given the data $\{y, C\}$. A transformation tree uses an adaptive local likelihood maximization approach for that purpose. Specifically, it selects $\hat{\theta}$ as

$$\hat{\theta}^N(c) = \arg\max_{\theta \epsilon \Theta} \sum_{i=1}^{N} w_i(c)\, \ell_i(\theta), \tag{5}$$

where $i \in \{1, \ldots, N\}$ denotes each observation in the dataset, and $\ell_i(\theta)$ denotes the log-likelihood contribution of i, when the parameters are given by θ. The recursive binary splitting process that creates a transformation tree is implemented by choosing weights:

$$w_i(c) = \sum_{b=1}^{B} I(c \in \mathcal{B}_b \wedge c_i \in \mathcal{B}_b). \tag{6}$$

The indicator function takes value 1 when observation i is sufficiently 'close' to c, so the weights in equation (6) simply count the number of observations in each bin $\mathcal{B}_b$. At the terminal nodes, $\mathcal{B}_b$ corresponds to a type, so the maximization process in equations (5) and (6) allocates each observation to a type, and sums the local likelihood functions across types. The type partition and the parameter vector θ are chosen so as to maximize that weighted sum of likelihoods. That is, given the available data $\{\boldsymbol{y}, \boldsymbol{C}\}$ and the recursive splitting approach to weights, the likeliest set of types and income distributions conditional on type is that given by $F\left(\tilde{y}_{qc}, \hat{\theta}^N(c)\right)$. The final type partition is thus the result of the interaction of the selected circumstances, and our prediction function under this method is given by:

$$\hat{y}_T = \hat{f}_T(c) = \frac{\mu}{\mu_q}\tilde{y}_{qc}, \quad \text{where } \tilde{y}_{qc} = F^{-1}\left(q, \hat{\theta}^N(c)\right). \tag{7}$$

The transformation tree estimate of absolute inherited inequality is then $I_n^A(y, \mathbf{c}, f_T) = I\left(\hat{y}_T\right)$, while the relative measure is analogously given by.

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$$I_n^R(y, \boldsymbol{c}, f_T) = \frac{I(\hat{y}_T)}{I(y)}. \tag{8}$$

Details of how the likelihood maximization is implemented (using Bernstein polynomials to fit the conditional distribution functions at each node) are given in Appendix Subsubsection A.1.1. In practice, the process can be summarized by the following seven-step algorithm.

1. Set a confidence level $(1 - \alpha)$ and a minimum size for final nodes (n_{min}).
2. Choose a polynomial order (M).
3. Estimate the unconditional distribution function with a Bernstein polynomial of order M.
4. Test the null hypothesis of polynomial parameter stability for all possible partitions based on each element of the circumstance vector $\boldsymbol{c}$, and store p-values.
5. If, for all $\boldsymbol{c}$ and each possible partition, either the Bonferroni-adjusted p-value is greater than α or $n_c < n_{min}$, then exit the algorithm.
6. Otherwise, choose the variable and the splitting value producing the smallest p-value to obtain two subgroups.
7. Repeat steps 4–6 for the resulting subgroups, until exiting everywhere.

In our application, we follow statistical convention and set α to 0.01 and n_{min} to 1% of the sample size. Then we choose M, the order of the Bernstein polynomial. The selection of M is not as simple as that of α, because how well a polynomial of a certain order interpolates the distribution is intrinsically data-dependent. An order too small might result in a poor approximation of the distribution, while too high an order would translate into a loss of degrees of freedom and high computational costs.[7]

To find an appropriate order, we tune the algorithm by estimating the out-of-sample log-likelihood, after a fivefold cross-validation, for several order values of the Bernstein polynomial (ranging between 2 and 10). We select the lowest order for which the relative improvement of the log-likelihood that would be obtained by estimating an additional parameter is smaller than 0.1%.[8] In step 3, an unconditional CDF for our sample is thus estimated with a Bernstein polynomial of order 8.

The key step is then step 4, where an M-fluctuation test is performed to detect instability of the parameters in the conditional distribution functions across potential types (see Appendix Subsubsection A.1.1). To intuitively illustrate this key test, Appendix Subsubsection A.1.2 provides a simple example of the procedure, using made-up data. Further details can be found in Hothorn and Zeileis (2021) and Kopf *et al.* (2013). After following steps 4–7, we obtain an estimated transformation tree, and from that tree several outputs that are described in Section 5. Before presenting those results, we briefly describe our datasets in Section 4.

4 | DATA

We apply this method to four countries: China, India, the USA and South Africa. These countries were selected for their relative importance in different parts of the global economy, and because they represent substantial heterogeneity in both the structure of inequality and the availability of data. For all countries, our samples comprise all adult individuals (aged over 18) observed in nationally representative surveys. The outcome of interest is equivalized disposable household income, using the square root equivalence scale (Buhmann *et al.* 1988; OECD 2013). We also age-adjust the income to account, at least in part, for lifecycle dynamics. The adjustment consists of regressing our income variable on age and age squared, and using the sum of the constant and

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the individual residual from that regression as the adjusted outcome variable (see, for example, Palomino *et al.* 2022).

We select a set of circumstance variables available in each survey to estimate inherited inequality. Balancing the desirability both to include the most relevant circumstances and to preserve a certain degree of comparability across countries, we selected the following seven variables: biological sex, place of birth, mother's and father's education, mother's and father's occupations, and an 'ethnicity' variable, which includes religion and a coarse caste classification in India.[9] Even for this relatively limited set of inherited characteristics, only two surveys (for China and the USA) contain information on all seven. Birthplace is missing for South Africa, and mother's and father's occupations are missing for India; comparisons should be interpreted accordingly.

For China, we use the China Family Panel Studies (CFPS), carried out by the Institute of Social Science Survey of Peking University every two years from 2010 to 2018 with a sample size of individuals with complete information ranging between 16,000 and 21,000 observations. The CFPS have been used already to study aspects of the intergenerational persistence of income and IOp in China (e.g. Fan *et al.* 2021; Emran *et al.* 2023). The dataset contains data on important inherited characteristics. We include: sex; ethnicity, classified into 11 categories (Han, Mongol, Hui, Tibetan, Miao, Yi, Zhuang, Bouyei, Korean, Manchu, Other); 24 birth area categories[10]; mother's and father's education in eight categories[11]; and mother's and father's occupations (10 ISCO categories plus one category for unemployment).

For India, we use the India Human Development Survey (IHDS), conducted by researchers from the University of Maryland and the National Council of Applied Economic Research. These are large representative samples of the 2005 and 2012 populations, each containing over 100,000 observations, resulting in analysis samples of 78,000 and 98,000 complete observations, respectively. The IHDS has previously been used to study intergenerational mobility and IOp in India (e.g. Asher *et al.* 2024; Kundu and Lefranc 2020). Circumstance variables included are: six castes/religions (Forward caste, Other Backward castes, Dalit, Adivasi, Muslim, Christian/Sikh/Jain); 23 birth areas[12]; and mother's and father's education in seven categories.[13]

For the USA, we employ the Panel Study of Income Dynamics (PSID), a well-known data source for researchers interested in the intergenerational transmission of income and status, as it is the longest-running longitudinal household survey in the world. The survey is currently managed by the University of Michigan, and has been used in many studies of intergenerational mobility and IOp in the USA(e.g. Mazumder 2018; Pistolesi 2009). We use every other wave (even years) between 1970 and 2018.

The PSID includes data related to employment, income, wealth, expenditures, and a number of other background characteristics that could be used as circumstance variables. However, to preserve a modicum of comparability with the other three countries, we restrict inclusion to the seven variables listed above. Ethnicity is described in six possible categories (White, Black, American Indian/Aleut/Eskimo, Asian or Pacific Islander, Hispanic, Other). The area of birth is also classified into six categories (Northeast, Northwest, South, West, Alaska/Hawaii, Foreign country). Mother's and father's occupations are coded in a variable based on ISCO codes (High includes ISCO 1, 2, 3; Medium includes ISCO 4, 5, 6; Low includes ISCO 7, 8, 9, 0).[14] Finally, mother's and father's education is recoded in eight categories (0–5 grades, 6–8 grades, 9–11 grades, High school, 12 grades and non-academic training, College—no degree, College—degree, Advanced college).

For South Africa, we rely on the National Income Dynamics Study (NIDS 1–5), carried out by the Southern Africa Labour and Development Research Unit. The NIDS is a longitudinal survey, collected in 2008, 2010/11, 2012, 2014/15 and 2017. It is an interesting dataset for studying the inheritance of inequality because it is a reliable and extensive source of information about incomes and circumstances for arguably the world's most unequal country. Inequality of opportunity and mobility have already been analysed in South Africa using the NIDS, for example by Piraino (2015) and Brunori *et al.* (2021). The circumstance variables that we include in the

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Table 1 Descriptive income statistics for the most recent waves in each country.

Country	Year	Mean	Gini	MLD	Top 1%	Top 10%	Bottom 40%
China	2018	9998	0.497	0.459	0.137	0.400	0.129
India	2012	3196	0.527	0.518	0.123	0.439	0.089
South Africa	2017	13,429	0.610	0.690	0.157	0.444	0.113
USA	2018	48,420	0.389	0.301	0.063	0.290	0.179

Notes: Income units are in 2017 US dollars at PPP exchange rates. MLD stands for mean log deviation. The last three columns represent the shares of income received by the top 1%, the top 10%, and the bottom 40% in the income distribution.
Source: CFPS (2018), IHDS (2012), NIDS (2017) and PSID (2018).

analysis are: ethnicity (African, Asian/Indian, Coloured, White); fathers' and mothers' education (13 categories, ranging from 'Not educated' to 'Grade 12 or more'); and fathers' and mothers' occupations (11 categories, 10 associated to the 1-digit ISCO, and one extra including other categories, such as out of the labour force, deceased or other unclassified occupations).

In all four countries, the final sample used for our analysis includes only complete observations, in the sense that information is not missing for any of the outcome or circumstance variables described above. Of course, item non-response can be a serious issue in data containing retrospective information on a respondent's parents. The process of dropping observations with incomplete information does reduce our sample sizes, and may do so in a selected way. While we cannot rule out sample selection biases, for each country and wave we examined the pattern of missing information and calculated the differences between average income (and inequality) when including and when excluding observations with missing circumstances. Results do not seem particularly alarming.

Table 1 shows some basic descriptive outcome statistics for the analysis samples of the most recent survey used. The four countries differ both in their level of development and in the nature of their inequality. The USA combines high income levels with a moderate level of inequality (at least relative to this group of countries). In contrast, South Africa, despite mid-level average incomes, is marked by high inequality, with a Gini coefficient exceeding 0.6. India and China, while showing similar overall levels of inequality, differ in the structure of that inequality. In China, income is more concentrated at the top: the top 1% accounts for 14% of total income, compared to 12% in India. However, India faces a more pressing issue at the lower end of the distribution. The bottom 40% of the population in India receives just 9% of total income, a figure that is not only lower than China's 13%, but also below that of South Africa. Appendix Table A1 contains the same set of summary descriptive statistics for all earlier waves.

5 | RESULTS: INHERITED INEQUALITY IN CHINA, INDIA, SOUTH AFRICA AND THE USA

To illustrate the full potential of the proposed method, Subsection 5.1 presents transformation trees and their associated visualization tools, illustrating both the structure of the partition of types as a sequence of splits, and the parametrization of heterogeneous CDFs. Subsection 5.2 then turns to the distribution of predicted incomes $\widehat{y}_T$ (defined by equation (7)), which capture the inherited component (i.e. predicted by circumstances) of total incomes. We illustrate how inherited inequality—inequality in the inherited component of income—can be compared with total inequality using Lorenz curves, and how it can be aggregated into a scalar measure. Finally, Subsection 5.3 presents two approaches to assessing the average and marginal predictive contributions of different circumstances, leveraging information from hundreds of trees.

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Table 2 Inherited inequality results for the most recent waves.

Country	Year (1)	Gini ($\hat{y}_T$) (2)	IR_T (Gini) (3)	MLD ($\hat{y}_T$) (4)	IR_T (MLD) (5)	Top 1% (6)	Top 10% (7)	Bottom 40% (8)
China	2018	0.292	58.8%	0.172	37.5%	0.079	0.270	0.245
India	2012	0.327	62.0%	0.207	40.0%	0.040	0.278	0.168
South Africa	2017	0.495	81.1%	0.413	59.9%	0.039	0.313	0.209
USA	2018	0.141	36.3%	0.052	17.3%	0.017	0.148	0.372

Notes: The last three columns present the shares of $\hat{y}_T$ received by the top 1%, the top 10%, and the bottom 40% in the predicted distribution.
Source: CFPS (2018), IHDS (2012), NIDS (2017) and PSID (2018).

5.1 | Visualization of type structure and inequality using transformation trees

Applying the algorithm outlined in Section 3 on the data described in Section 4, with the key stopping rule parameters set to $\alpha = 0.01$ and $n_{min} = 1\%$ of the sample, yields the transformation trees shown as Figures OA1–OA4 in the Online Appendix. The nodes resulting from those trees are used to obtain $\hat{y}_T$ and to measure inherited inequality (estimates described in Table 2 below). However, as these full trees are fairly deep and complex, Figure 1 shows a modified version for the USA, with the corresponding tree pruned to a maximum depth of four levels to make the output more readable.[15] Appendix Figures A6–A8 display the analogous pruned trees for China, India and South Africa.

The splitting process generated by the algorithm should be read from left to right. The first split in Figure 1 divides the population into a group consisting of around one-third of the sample, whose fathers had at least some college education (node 15, categories 6, 7 and 8), and the rest of the sample (node 2). As we move to the right, other circumstances further partition the population following the algorithm, until the final nodes—types—are reached. An interesting symmetry emerges straight away: ethnicity appears as the second splitting circumstance in both subtrees, producing identical splits (into nodes 16 and 19 for those with more educated parents, and nodes 3 and 8 for the remainder). In both cases, the categories Black and American Indian/Aleut/Eskimo cluster together in the poorer sub-branch, while all other ethnic groups fall into the more affluent sub-branch. Subsequent splits are determined by the father's education and occupation, the mother's education, birth area and sex. At level four, where the tree is trimmed, we find a partition into ten types, with terminal nodes showing their parametrically estimated density functions, indicating the population share accounted for by each type, and its mean income as a share of the overall mean.[16]

In terms of the model selection challenge discussed in Section 3, the algorithm partitioned the population into these ten groups (and fit CDFs to them) so as to maximize the likelihood of fitting the data, under the restrictions $f \in \mathcal{F}_\tau$, with $\mathcal{F}_\tau$ being the class of recursive binary transformation tree estimators. The partition can be thought of as the result of interacting, in a linear regression setting, various dummy variables defined over the circumstances. Type 10, for example, which is the poorest type in terms of expected income (54% of the national average) and comprises 13% of the population, consists of black or indigenous women whose fathers never went to college and whose mothers worked in specific occupations. This corresponds to the interaction of variables that can be written as

$$type_{10} = \mathbf{1}_{race=Black\ or\ American\ Indian/Aleut/Eskimo} \times \mathbf{1}_{father\ education=below\ college} \times \mathbf{1}_{sex=female} \times \mathbf{1}_{mother\ occupation=ISCO\ category\ 1\ or\ 2}.$$

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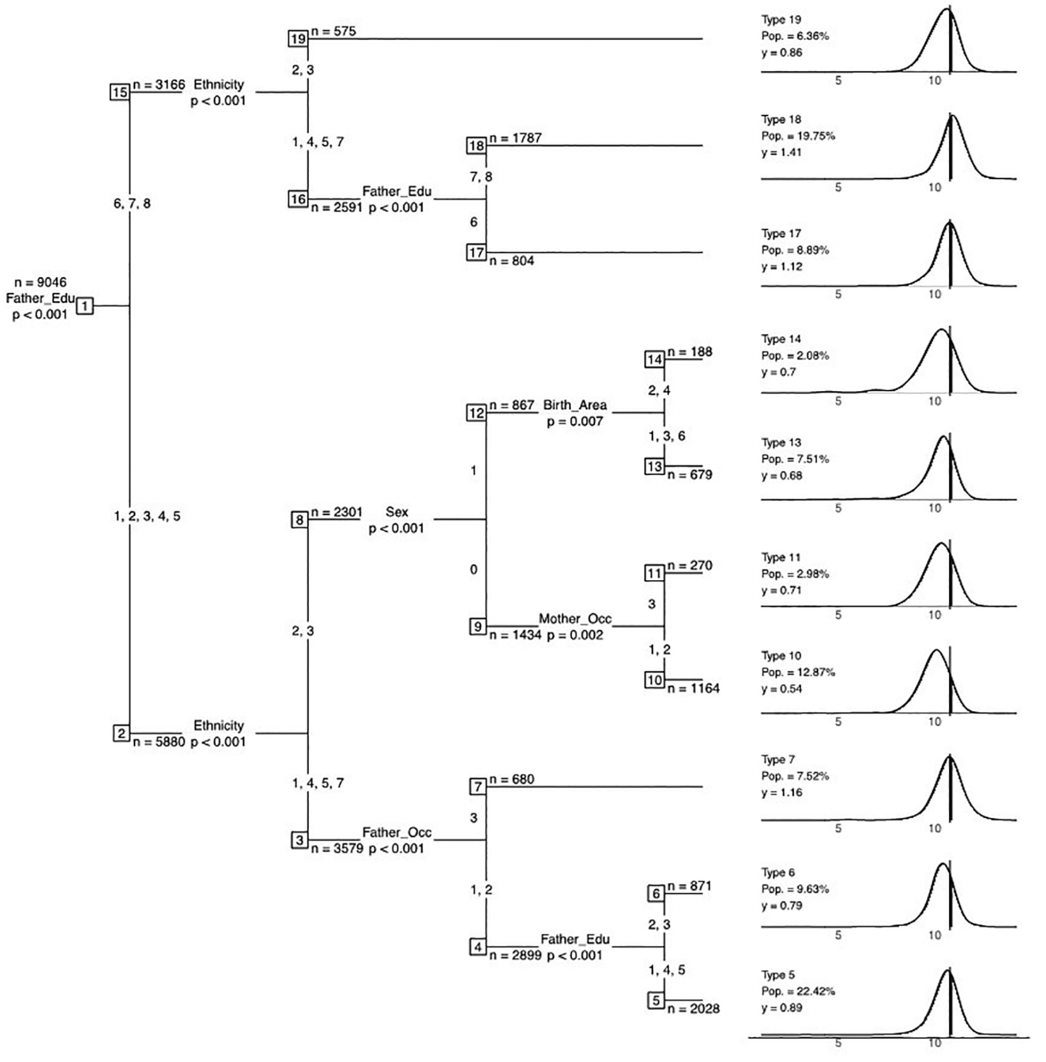

Figure 1 Transformation tree for the USA (2018). *Notes*: Splitting nodes show their sample size and the p-value associated with the split. Circumstance categories are Gender (0 Male, 1 Female), Ethnicity (1 White, 2 Black, 3 American Indian/Aleut/Eskimo, 4 Asian/Pacific Islander, 5 Hispanic, 7 Other), Region of upbringing (1 Northeast, 2 North Central, 3 South, 4 West, 5 Alaska/Hawaii, 6 Foreign country), Parents' education (1 for 0–5 grades, 2 for 6–8 grades, 3 for 9–11 grades, 4 for High school, 5 for 12+ grades + non-academic training, 6 for Some college, 7 for College degree, 8 for Advanced college degree), and Parents' occupations (ISCO) (1 Basic, 2 Middle, 3 High). The panels on the right display the log-density of type-specific incomes. The labels indicate the share of the population that each type represents (Pop.) and their average income relative to the overall sample mean ($y = 1$), which is also depicted as a vertical black line in the log-density plot. *Source*: Own elaboration from the PSID (2018).

Type 18, which is the richest type and includes 20% of the sample, consists of White, Hispanic and Asian people whose fathers are college graduates, corresponding to the interaction

$$type_{18} = \mathbf{1}_{race=White,Hispanic,Asian} \times \mathbf{1}_{father\ education=college\ graduate\ or\ more}.$$

And so on.

For China (see Appendix Figure A6), the four-level tree yields a 13-type partition, with incomes ranging from 47% to 253% of the mean income. The birth area variable plays a critical

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role in the transformation tree: it is the first splitting factor, and it also appears in five additional splitting nodes, highlighting the importance of the geography of birth for the distribution of life chances in China. For example, the two most advantaged types consist exclusively of individuals born in Shanghai. Other influential variables include parental occupation, education, and, to a lesser extent, ethnicity (the worst-off type contains only Mongol and Yi individuals).[17]

For India (see Appendix Figure A7), the four-level tree yields fifteen types, with incomes ranging from 47% (type 15) to 374% (type 29) of the mean. Father's education determines the first split in the transformation tree, followed by mother's education. The structure of the tree indicates a dominant role for the interaction between these two variables. The more affluent group, whose expected income is nearly four times the sample mean, is composed of individuals whose parents are both at the top of the educational distribution. Additionally, birth area and ethnicity (which here consists of caste and religious identities) also contribute substantially to predicting the shape of the income distribution, which is heterogeneous not only in terms of its mean, but also in its higher-order moments. While the poorest types tend to follow a log-normal distribution, the richer types exhibit distributions skewed to the right, suggesting greater variability, and possibly some downward risk (in the long left tails) for these higher-income groups. These insights into the shape of each type's distribution are one benefit of our approach to measuring inherited inequality using transformation trees.

Finally, for South Africa (see Appendix Figure A8), the pruned tree has nine nodes or types. Unsurprisingly, its structure is determined primarily by race. The white population, which is exclusively concentrated in types 16 and 17, displays income distributions so markedly shifted to the right that it appears as though they might have been drawn from a different country. Among the non-white groups, those including Asian and Coloured individuals have a higher expected income and tend to follow a log-normal distribution. In contrast, groups composed of African individuals exhibit a distribution with a density mass skewed to the left.

Besides the parametrized density functions shown to the right of the trees in Figure 1, type distributions can also be visualized as CDFs. Figure 2 shows both the empirical CDFs (ECDFs) for each type as solid lines, and the corresponding predicted CDFs, $F\left(\tilde{y}_{qc}, \hat{\theta}^N(c)\right)$, generated by the Bernstein polynomials, as dashed lines.[18]

These CDF plots provide a striking visual depiction of the structure and extent of inherited inequality in each country. Compared to the USA, type distributions are much further apart in the three developing countries in our sample, particularly South Africa, where the two White types (16 and 17) stochastically dominate all others by a large margin. Consider also types 12 and 9 in South Africa, which can be characterized as intermediate types with expected incomes around 90% of the population mean. Although they differ in the interaction between parental education and occupation, the most salient distinction is their racial composition: type 12 comprises only Africans, whereas type 9 is made up of Coloured respondents. The income distribution for type 9 is substantially more compressed, while type 12 exhibits greater dispersion. The implication is that membership in type 9 yields a relative advantage over type 12 at the lower tail of the conditional income distribution, while high-performing individuals in type 12 outperform high achievers in type 9. Although these two types have similar means, these higher-moment differences are material features of what it means to have their inherited circumstances. Similar crossings can be observed in all four countries, revealing that types differ not only in terms of their means, but also in their higher-order moments.

5.2 | From trees to Lorenz curves and scalar measures of inherited inequality

As discussed, Figures 1 and 2, and Appendix Figures A6–A8, draw on four-level pruned trees. To compute the degree of inherited inequalities we rely on the full, deeper trees shown in

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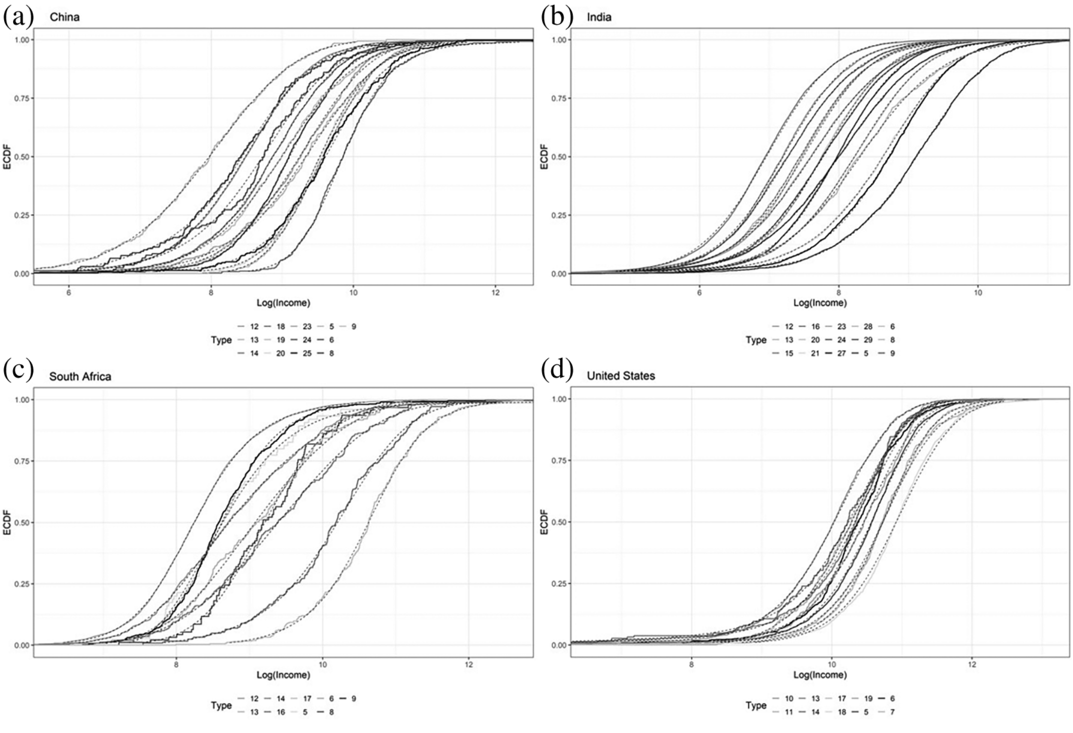

Figure 2 CDFs of the type partition (pruned trees) for the most recent waves. *Notes*: Here we present the empirical and predicted CDFs of the types obtained from the pruned transformation trees displayed in Figure 1 (USA), Appendix Figure A6 (China), Appendix Figure A7 (India) and Appendix Figure A8 (South Africa). Solid lines represent the ECDFs for each type, and are labelled consistently with the corresponding types in the trees. Dashed lines represent the corresponding CDF predicted with the Bernstein polynomial. *Source*: CFPS (2018), IHDS (2012), NIDS (2017) and PSID (2018).

Figures OA1–OA4 in the Online Appendix. Those trees generate finer types, with their own income predictions $\tilde{y}_{qc}$. These are then adjusted for differences across quantile means as in equation (7) to yield the distribution of predicted incomes $\hat{y}_T$, which is used for computing our proposed absolute and relative measures of inherited inequality or IOp (equation (8)).

We can also define a Lorenz curve of these predicted incomes, $L(\hat{y}_T)$, which we call the inherited component Lorenz curve (ICLC), since that is what the incomes predicted by circumstances capture. Figure 3 displays both ICLCs and regular Lorenz curves $L(y)$ for the most recent surveys from each country. The ICLCs dominate the regular Lorenz curves, since the counterfactual distribution contains only a fraction of the original inequality, having excluded all inequality within types. The area between $L(\hat{y}_T)$ and the line of equality captures the extent of inherited inequality in each society. The dashed lines at certain quantiles indicate the shares of total income (from $L(y)$) and of its inherited component (from $L(\hat{y}_T)$) held by the bottom 40%, 90% and 99% of the population, as reported in Table 2. A comparison reveals that for South Africa, the two curves are strikingly close: the distribution of predicted incomes closely tracks that of actual incomes; moreover, at the ninth decile, they both have a point of increasing curvature. This suggests that the advantages observed at the tops of the distributions are substantial and are almost entirely attributable to inherited circumstances.

To go from the ICLCs in Figure 3 to scalar measures of inherited inequality we choose two different indices, namely the Gini coefficient and the mean log deviation (MLD). Values for income inequality were presented in Table 1, and Table 2 contains the corresponding inherited

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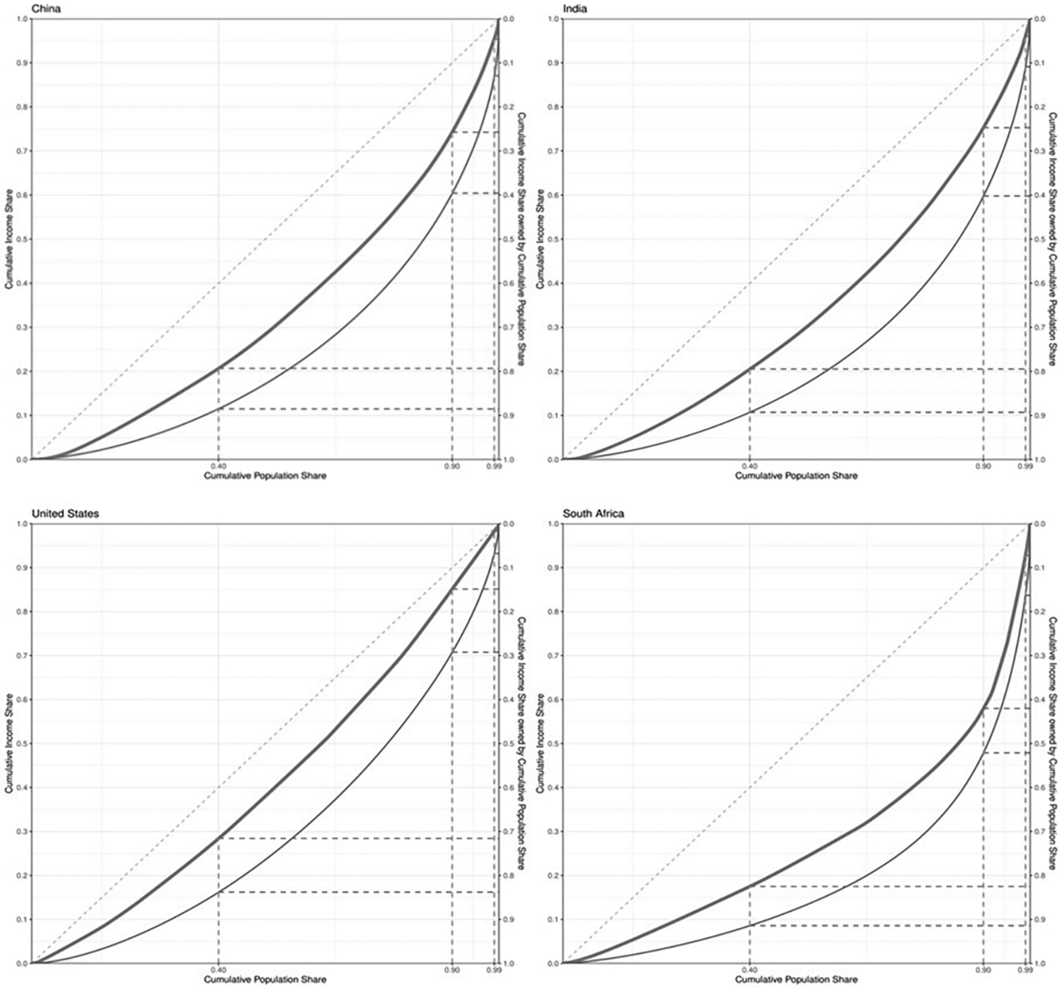

Figure 3 Lorenz curves for total income and the inherited component of income in four countries. *Notes*: The thick solid line corresponds to the ICLC, while the light solid line corresponds to the regular Lorenz curve for total income. The dashed lines correspond to the income (Table 1) and inherited component (Table 2) shares received by the bottom 40%, top 10% and top 1%. *Source*: CFPS (2018), IHDS (2012), NIDS (2017) and PSID (2018).

inequality measures, once again for the latest available survey waves, while Appendix Table A2 display the results for all other waves. Although we report both Gini coefficients and MLDs, we focus the subsequent discussion on the Gini estimates.[19] For both indices, we report the absolute measure of inherited inequality, $I_n^A(y, \boldsymbol{c}, f_T) = I\left(\widehat{y}_T\right)$, as well as the relative measure $I_n^R(y, \boldsymbol{c}, f_T) = I\left(\widehat{y}_T\right)/I(y)$, denoted as IR_T. The last three columns of the tables report the shares of the inherited component of income accruing to the top 1%, top 10% and bottom 40% of the population.

For these particular years, the inherited Gini coefficient ranges from 0.141 in the USA to 0.495 in South Africa. The latter is a remarkable number: the inherited inequality Gini coefficient for South Africa is higher than the overall income Gini coefficient for the USA, and almost as high as total inequality in China. Indeed, the inherited component accounts for a remarkable 81% of the (very high) income inequality in South Africa. India has the second-highest level of inherited inequality, with Gini coefficient 0.33, accounting for 62% of overall inequality, with China not far behind. However, the shape of the distribution of the inherited component

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differs between those two countries, mirroring what we have already seen for the distribution of incomes. In India, the top 1% receives 4% of the inherited component of income, half of the share in China, where the top 1% holds 8%. Yet India's top 10% captures about the same as in China, while the bottom 40% receives significantly less (17% compared to 25% in China), even than in South Africa (21%). This suggests that although India exhibits less concentration at the very top of the distribution of the inherited component of incomes as compared to China, it has a much larger share of the population that is extremely deprived in that dimension. Appendix Figure A9 presents the time trends of inherited inequalities for all four countries since 2000, revealing considerable stability in the USA during the 2000–2018 period, while values rose both in India (between 2005 and 2012) and China (particularly since 2014). South Africa followed a U-shaped pattern, with inherited inequality declining until 2015, then rising again in 2017.

It is hard to compare our main results with previous studies, which have employed various different statistical approaches and often used different samples and income definitions. For example, in the case of the USA, Pistolesi (2009) reports a similar level of IOp in terms of MLD for the year 2000 to what we obtain. However, his analysis focuses on earnings and is restricted to working males, which limits comparability. In most other cases, previous estimates are generally based on *ad hoc* type partitions, and these tend to be lower—often much lower—than ours. For India, Kundu and Lefranc (2020) estimate that IOp in 2012 ranges from 8% to 39%, depending on the set of regressors used in a parametric model. Their estimate using conditional inference regression trees is 32%, which compares to our relative inherited inequality estimate of 62%. For China, Wu (2018), using the same data source and inequality index (Gini) as our study, reports relative IOp levels between 30% and 40% for the years 2010 and 2012, whereas we observe relative levels of 53% in 2010, and 42% in 2012. Finally, Piraino (2015) uses two econometric methods to estimate IOp in gross employment earnings in South Africa, using up to 54 Roemerian types. Using data for male workers in 2008 and 2012, he finds IOp shares ranging from 17% to 24% of total inequality, as measured by MLD. These estimates compare to our MLD-based estimate of 57% in 2012. Once again, although comparisons are hampered by differences in samples and/or income definitions, most previous IOp estimates for our sample of countries are considerably lower than ours.

5.3 | The role of individual circumstances: average and marginal contributions

Because the prediction function in equation (7) is highly non-linear in circumstances, any assessment of the relative contribution of individual circumstances to inequality in predicted incomes, $I\left(\hat{y}_T\right)$, cannot rely on marginal effects. As in other cases in inequality analysis, the decomposition method most suitable to our application is the Shapley–Shorrocks decomposition (Shapley 1953; Shorrocks 2013). Intuitively, this decomposition computes the total contribution of a particular circumstance variable c_k to predicted inequality as the average reduction in the latter when c_k is omitted from the prediction, with the average taken across all possible combinations of circumstances that originally include c_k (see Shorrocks 2013). A description of the algorithm used to compute the decomposition helps to clarify its logic.

1. Draw a subsample of the full sample.[20]
2. Estimate inherited inequality in this subsample, as described in Section 3 (equations (7) and (8)), but setting $\alpha = 1$ and $n_{min} = 0$.
3. Further, estimate inherited inequality in the subsample for all possible permutation sequences that eliminate circumstance c_k. This elimination is performed by replacing c_k with a constant vector $\mathbf{1}$.
4. Estimate a tree and inherited inequality after each elimination sequence, and store results.

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Table 3 Shapley value decomposition of inherited inequalities (%) for the most recent waves.

Circumstances	China	India	South Africa	USA
Birth area	15.41	20.87	—	14.25
Ethnicity	8.65	16.32	32.41	15.24
Father's education	10.46	27.74	16.43	19.17
Father's occupation	16.32	—	13.78	15.42
Mother's education	17.26	27.89	16.07	16.5
Mother's occupation	29.53	—	16.06	12.29
Sex	2.36	7.17	5.25	7.13

Notes: Values represent the relative contribution (%) of circumstances to the inherited inequality estimates reported in column (2) of Table 2. The values within a column sum to 100%. Missing values correspond to circumstances that are not available in those datasets. *Source*: CFPS (2018), IHDS (2012), NIDS (2017) and PSID (2018).

5. Average the estimates of inherited inequality across all permutation sequences. The difference between overall inherited inequality and this average is the specific contribution of c_k.
6. Repeat steps 1–5 z times, to account for different potential data-generating processes. In our case, we set $z = 100$.
7. Estimate the contribution of c_k to inherited inequality as the average contribution across these z repetitions.
8. Repeat the algorithm for each c_k, $k \in \{1, \dots, K\}$.

This algorithm grows trees on subsamples of the initial population, permitting each tree to attain significant depth. These two adjustments enable all circumstances with predictive power to contribute to defining the partition of types, at least in some iterations, making the assessment of the relative contribution of each circumstance more robust to the typical problem of the variance of estimates based on a single tree. Table 3 presents the results of the Shapley–Shorrocks decomposition for our four countries and the seven circumstance variables available in the most recent wave. The predictive power of inherited circumstances should be understood as conditional on the prevailing economic and institutional environment. Since our estimates cannot control for other determinants of inequality that may be correlated with circumstances, the analysis does not identify causal effects and is purely descriptive. Results are presented as percentage shares of the inherited inequality Gini coefficients reported in Table 2.

Notice that due to the unobservability of parental occupation in India and area of birth in South Africa, results are fully comparable only for China and the USA.[21] Looking across those two countries, it is interesting that despite the larger number of ethnic groups in China, the Shapley value for race in the USA is twice as large. While father's occupation and mother's education have similar values in both countries, father's education appears significantly more important in the USA, whereas mother's occupation plays a larger role in China. The influence of sex also appears to be much more pronounced in the USA.

When including India and South Africa in the comparison, the dominant role of race emerges clearly in South Africa: it is comparable to the combined effect of caste and area of birth in India. Interestingly, the Shapley value for parents' education in India is not too far from the sum of the values for both occupation and education of the parent in South Africa, suggesting that the presence of unobservable circumstances may inflate the Shapley value of observable and correlated circumstances. Regarding the role of sex, it is important to recall that our analysis is based on equivalized household income. Therefore we expect sex to play a significant role only in contexts where single-parent households are not uncommon, or where differences in household gender composition are systematically correlated with total household incomes. This is the case in India and the USA. In contrast, the difference is smaller in our sample for China, where female-headed

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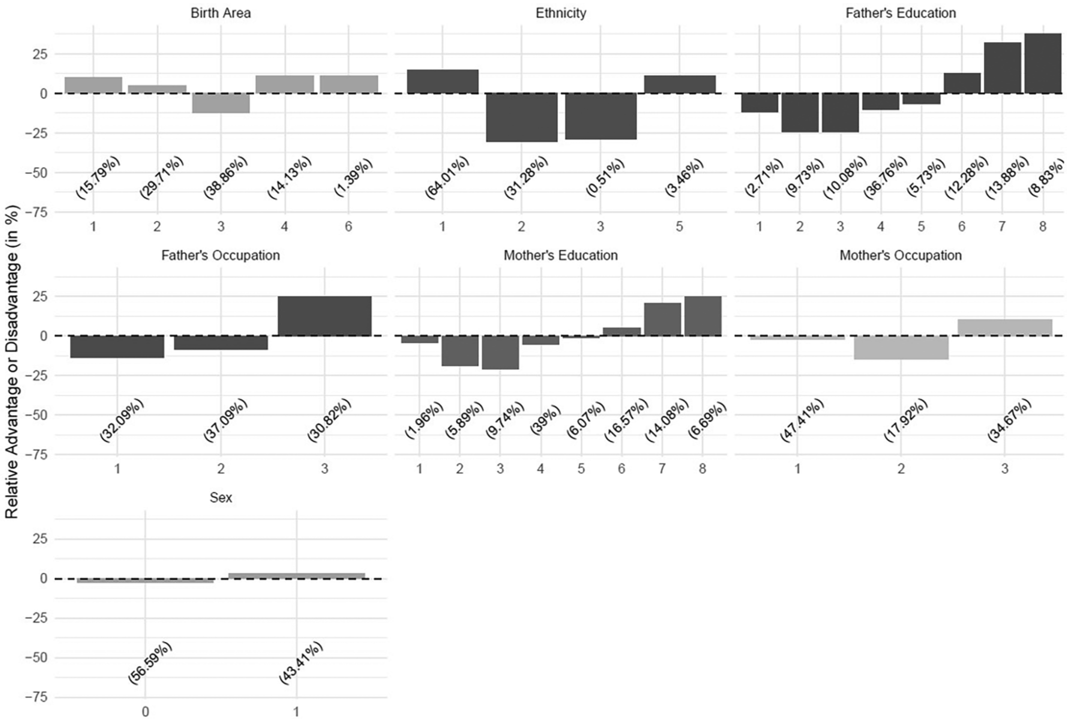

Figure 4 Marginal effects of circumstances on the inherited component of income in the USA. *Notes*: Values on the vertical axis represent the marginal effects associated with each category, computed as described in the text. Circumstance categories are Gender (0 Male, 1 Female), Ethnicity (1 White, 2 Black, 3 American Indian/Aleut/Eskimo, 4 Asian/Pacific Islander, 5 Hispanic, 7 Other), Region of upbringing (1 Northeast, 2 North Central, 3 South, 4 West, 5 Alaska/Hawaii, 6 Foreign country), Parents' education (1 for 0–5 grades, 2 for 6–8 grades, 3 for 9–11 grades, 4 for High school, 5 for 12+ grades + non-academic training, 6 for Some college, 7 for College degree, 8 for Advanced college degree), and Parents' occupations (ISCO) (1 Basic, 2 Middle, 3 High). Numbers in parentheses denote population shares in each category. We are not showing categories populated by less than 0.5% of the sample size. *Source*: PSID (2018).

households earn about 95% of what their male-headed counterparts earn. Once again, in keeping with the measurement-using-prediction spirit of our analysis, these decompositions are purely descriptive.

Moreover, when commenting on the role of circumstances in predicting the conditional distribution of income, we should consider that Shapley values measure the reduction in predicted inequality when a specific circumstance is removed from the analysis. This value is influenced by the distribution of the circumstance itself. For example, in a society where most individuals are Black and only a few are White, removing the race variable will not significantly reduce the model's explanatory power. This is because for the majority, the conditional distribution closely aligns with the unconditional one, and only for a small subset does race influence the income distribution. However, from the point of view of the minority, that characteristic may matter a great deal. In other words: while the average contribution of individual circumstances contains valuable information, so would an estimate of the marginal importance of belonging to a specific circumstance category.

We therefore complement Shapley values by estimating the marginal effect (on predicted incomes) of having a specific characteristic, for example, being White in the USA. We do this by regressing individual predicted incomes $\widehat{y}_T$ on dummy variables representing each category for a given circumstance. These regressions are run separately for each circumstance and without a constant term. The marginal effect of a characteristic is then the estimated coefficient on the

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dummy variable for the corresponding category, normalized by the average of predicted incomes. This exercise is similar to the estimation of partial dependence plots, which are frequently used in machine learning to complement decomposition techniques such as Shapley values.

Figure 4 presents the marginal effects for the most recent wave for the USA. Both the signs and magnitudes of the effects are broadly consistent with expectations.[22] The positive effect of being White is clearly visible in the middle panel at the top, and it is associated with an opportunity premium of 14%. A substantial 'college premium' can also be observed for both fathers' and mothers' education (category 6 and above in the respective parental education panels). The only area of birth showing a large negative effect is the South, with a penalty of over 10% (the macro region includes Alabama, Arkansas, Delaware, District of Columbia, Oklahoma, South Carolina, Tennessee, Texas, Virginia and West Virginia).

Appendix Figures A10–A12 show analogous marginal effects for China, India and South Africa. Noteworthy findings include the strong positive effect of being born in certain regions of China—for instance, being born in Shanghai is estimated to contribute a marginal advantage of more than 150% to predicted incomes. In India, having well-educated parents is associated with significantly higher inherited components. This premium is particularly high for maternal education: for the small minority reporting a mother with education above the secondary level, the estimated marginal advantage exceeds 200%. In South Africa, a sizeable positive effect is observed for the small White community, with an estimated inherited component premium of 350%.

6 | CONCLUSIONS

The extent to which economic advantage is inherited from previous generations and shaped by predetermined circumstances is a matter of both positive and normative interest. Many, if not most, approaches to quantifying this phenomenon rely on prediction exercises, essentially assessing how well incomes can be predicted by predetermined circumstances such as parental income, biological sex, race, or other indicators of family background. We showed that many commonly used measures of intergenerational mobility and IOp can be written as functions of the ratio of inequality in these predicted incomes to inequality in observed current-generation incomes.

We then proposed a new approach for measuring inherited inequality that is sensitive to differences across the full conditional income distributions—rather than just the means—of subpopulations that share the same inherited characteristics—'types' in the IOp literature. This method, based on transformation trees (Hothorn and Zeileis 2021), represents an improvement over previous approaches to estimating inherited inequalities because it is designed to partition the population and estimate distribution functions optimally, in the face of the trade-off between a downward omitted-variable bias and an upward overfitting bias that is inherent to the model selection problem in this literature.

We applied this method to 36 representative household surveys from four large and systemically important countries, namely the USA (25 waves, between 1970 and 2018), China (five waves, between 2010 and 2018), India (two waves, 2005 and 2012) and South Africa (four waves, between 2008 and 2017). We found high absolute levels of inherited inequality, with Gini coefficients 0.14 in the USA, 0.29 in China, 0.33 in India, and 0.50 in South Africa in the latest available years. These correspond to substantial shares of total income inequality—36% for the USA, 59% for China, 62% for India, and 81% for South Africa—attesting to the heavy weight of inherited characteristics in predicting current economic success in all four countries, but particularly the three poorer ones.

We also estimated both average and marginal contributions of specific circumstances (and categories, in the marginal case) to overall inherited inequality. The relative importance of these circumstances varied substantially across countries, reflecting their different histories and socioeconomic structures. Race was unsurprisingly dominant in post-Apartheid South Africa,

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whereas area of birth was important in both India and China, where being born in cities such as Shanghai yields great advantage. But it was the occupation of one's mother that seemed to be the most descriptively important inherited characteristic in China, whereas the educational attainment of one's father played that role in the USA. Being Black in the USA commands a significant (25%) inherited component penalty.

Most of these insights into the extent and nature of the inheritance of inequality across generations, and of the distribution of opportunities across these four very different countries, were possible only through a comprehensive approach that incorporates many circumstance variables and looks beyond averages—and at differences across full conditional distributions—when assessing how predictive they are of observed living standards.

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ENDNOTES

[1] See Foster and Shneyerov (2000) and Ebert (2010) for discussions of why it might make sense to account for differences in the full distributions within groups—rather than just the means—when defining the between-group term of the decomposition.

[2] See Ferreira and Brunori (2024) for a more in-depth discussion of the concept of inherited inequality and of the relationship between intergenerational mobility, IOp and this broader concept.

[3] Formally: $\mu_q = (1/M)\sum_{m=1}^{M} F^{-1}(q|\boldsymbol{c}_m)$.

[4] The overall mean μ restores the distribution to its original scale, but since it is a constant across all observations, it has no effect on the inequality computation.

[5] The working paper version of this article (Brunori *et al.* 2026) contains a more detailed discussion of how our approach relates to the IOp literature.

[6] This bias is also a concern for estimates of intergenerational mobility if they are to be interpreted as measures of inherited inequality—except in the unlikely event that parental income is a sufficient statistic for all circumstances.

[7] The confidence level $(1-\alpha)$ and the order of the polynomial (M) interact in determining the depth of the tree, and thus the complexity of the final partition. For a given sample size, fixing a higher polynomial order implies using more degrees of freedom in each test, leading to a lower probability of rejecting the null hypothesis of equal distributions. Consequently, the resulting partition is more parsimonious. Similarly, for a given polynomial order and confidence level, a larger sample size results in a more detailed partition of types and likely a higher level of between-group inequality.

[8] Note that the order of the polynomial does not turn out to be a key determinant of the estimated level of inherited inequalities. Appendix Figure A5 shows, for the case of South Africa, the stability of the estimate when the selected Bernstein polynomial order varies.

[9] Given the choice of household income as the outcome of interest, the role of sex among the circumstances will necessarily be limited, since intra-household inequalities are ignored. The relative unimportance of sex in the analysis that follows (see Table 3 and Figure 4, for example) should therefore be treated cautiously. This does not imply that these countries provide equal opportunities to men and women in other dimensions. On the other hand, omitting sex as a circumstance variable would have caused us to miss non-negligible consequences of differences in household composition across countries.

[10] The birth areas are Hebei, Shanxi, Liaoning, Jilin, Heilongjiang, Shanghai, Jiangsu, Zhejiang, Anhui, Fujian, Jiangxi, Shandong, Henan, Hubei, Hunan, Guangdong, Guangxi Zhuang Autonomous Region, Sichuan, Guizhou, Yunnan, Shaanxi, Gansu, plus Not available and Other.

[11] The education categories are Illiterate/semi-literate, Primary school, Junior high school, Senior high school/secondary school/technical school/vocational senior school, 3-year college, 4-year college/bachelor's degree, Master's degree, Doctoral degree.

[12] The birth areas are Jammu and Kashmir, Himachal Pradesh, Punjab, Another state, Uttarakhand, Haryana, Delhi, Rajasthan, Uttar Pradesh, Bihar, Overseas, Northeast, West Bengal, Jharkhand, Orissa, Chhattisgarh, Madhya Pradesh, Gujarat, Maharashtra, Andhra Pradesh, Karnataka, Kerala, Tamil Nadu.

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[13] The education categories are None, Incomplete primary, Complete primary, Incomplete secondary, Complete secondary, Higher secondary, Post-secondary or higher.
[14] ISCO codes are grouped in this way following Hufe *et al.* (2022). Results are very similar if each ISCO occupational classification is entered separately.
[15] This is implemented through an additional stopping rule that prevents any observation from being involved in more than four splits. We remark that inherited inequalities estimates rely on the full trees, shown in Figures OA1–OA4 of the Online Appendix.
[16] Although we use income in levels to compute all our measures, we plot the density of log incomes for ease of visualization.
[17] The split at node 4 in the Chinese tree is worth a comment, as it groups parents with a master's degree together with the least educated category, in type 5. This illustrates a feature of the algorithm when there are very few observations in a particular category. When the algorithm uses a certain circumstance to divide the sample, it must place all individuals from the node that originates the split into either one subgroup or the other. If there are very few respondents who have a specific value for the characteristic in question—there are only four individuals in the entire sample reporting a father with a master's degree—then the assignment to the group can be almost random. Naturally, because this happens only when the number of individuals is very small, the consequences for inherited inequality estimation are minimal.
[18] Refer to the working paper version of this paper (Brunori *et al.* 2026) to see all figures in colour.
[19] As noted by Brunori *et al.* (2019), the Gini coefficient is more sensitive to the central parts of the distribution, where group means tend to cluster, rather than to the lower tail. In that sense, the Gini coefficient is better suited to studying IOp than the MLD. However, if one demands a scale-invariant relative measure that can be additively decomposed into between- and within-group components in a path-independent manner, then the only suitable option is the MLD.
[20] Following the convention often used in tree bagging procedures, we draw subsamples of 63.2% of the original sample size (see Hothorn *et al.* 2006).
[21] Even in this case, some might argue that comparability is limited by how the same circumstances are coded across countries. For example, in the USA, there are six racial categories, while in China, there are eleven ethnic groups. However, such classifications are linked to a country's structure of opportunity. We cannot and should not impose identical categories across countries, which differ in terms of their territories, cultural diversity and social structure. These aspects of a country's social organization are part and parcel of its opportunity distribution, and they should be reflected in the data used—without attempting to make them uniform. It is for the learning algorithm to select the most salient binary splits across categories in each case.
[22] We only show results for categories representing more than 0.5% of the sample.
[23] Ours is a different version of a similar example proposed by Kopf *et al.* (2013).

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APPENDIX A

A.1 Technical details of the transformation tree algorithm

A.1.1. The likelihood maximization using Bernstein polynomials

In practice, implementation of the likelihood maximization is facilitated by using a monotonic transformation function of y, namely $z = h(y)$, with $h'(y) > 0, \forall y$. Monotonicity ensures that $F(y) = F_z(h(y))$. We follow Hothorn and Zeileis (2021) in using Bernstein polynomials of order M to construct the transformation function: $h(y) = a(y)^T\theta$. Note that $a(y)$ is a polynomial of order M in y. The choice of M implies the choice of the dimension of the parameter vector, $P = M + 1$. The higher that order, the greater the flexibility with which $F\left(y_{qc}, \theta(c)\right)$ can be modelled, and the greater the degree to which differences in their higher moments affect the partition and the estimation. Bernstein polynomials are a particular application of this transformation function, in which.

$$a_M(y) = \frac{\left(\phi_{1,M+1}(y),\ \dots\ , \phi_{M+1,1}(y)\right)}{M+1}, \tag{A1}$$

where $\phi_{m,M}$ denotes the density of the beta distribution with parameters m and M. Using this particular vector for the polynomial in $h(y)$ implies a simple log-likelihood function that can be used for maximization:

$$\ell_i(\theta) = \log\left[f_z\left(a(y)^T\theta\right)\right] + \log\left(a(y)^T\theta\right). \tag{A2}$$

With this specific functional form for $\ell_i(\theta)$, all that is needed to solve equations (5) and (6)—and thus have the parameter estimates to model the conditional income distributions for all types in the tree terminal nodes—is the algorithm to split the sample into types. This proceeds sequentially. Start from the case when $w_i(c) = 1, \forall i$. This corresponds to no splits: all observations are in a single bin, and have the same weight in the log-likelihood maximization. The parameter estimates obtained under that assumption are the simple maximum likelihood estimates:

$$\hat{\theta}^N_{ML}(c) = \arg\max_{\theta \epsilon \Theta} \sum_{i=1}^{N} \ell_i(\theta). \tag{A3}$$

To decide whether or not a split can improve prediction, test the null hypothesis

$$H_0 : s\left(\hat{\theta}^N_{ML}|y\right) \perp C, \tag{A4}$$

where $s(\hat{\theta}|y)$ denotes the gradient contribution of observation i. For continuous distributions, the score contribution is simply the derivative of the log-density with respect to θ. Differentiating equation (A2), we obtain

$$s(\hat{\theta}|y) = a(y)\frac{f'_z\left(a(y)^T\theta\right)}{f_z\left(a(y)^T\theta\right)} + \frac{a'(y)}{a'(y)^T\theta}. \tag{A5}$$

There are a number of ways to test hypothesis (A4), and we follow Hothorn and Zeileis (2021) in using M-fluctuation tests. When these tests reject H_0, the algorithm implements a binary split in the circumstance x (an element of the vector c) that has the most significant association with the $P \times P$ score matrix, measured by the marginal multiplicity adjusted p-value (see Hothorn *et al.* 2006).

The algorithm is then repeated by testing hypotheses analogous to (A4) in each of the resulting cells, and so on recursively, until H_0 can no longer be rejected. At this point, the algorithm has

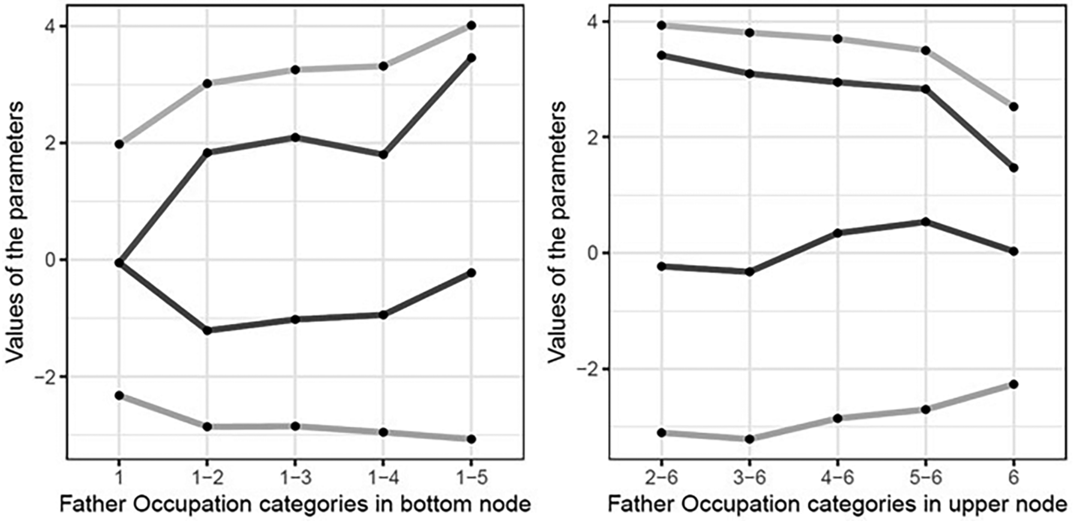

Figure A1 Values for the parameters of the Bernstein polynomial in each node. *Source*: Own elaboration on NIDS 5.

identified the optimal partition of the population into types: $\mathfrak{I} = \bigcup_{b=1,\ldots,B} \mathcal{B}_b$. Over this final partition, the likelihood function given by (A2) and the weights given by equation (6) are used to solve equation (5), yielding the final parameter vector $\widehat{\theta}^N(c)$, which fully characterizes the conditional distribution $F\big(y_{qc}, \theta(c)\big)$ in each type (terminal node) $\mathcal{B}_b$.

These parametric conditional distributions can then be inverted to yield the estimated type quantile functions $\tilde{y}_{qc} = F^{-1}\Big(q, \widehat{\theta}^N(c)\Big)$.

A.1.2. An illustration of the M-fluctuation test using made-up data

The algorithm employs an M-fluctuation test of parameter stability to determine node splits. Purely as an example, we show how the algorithm performs the type partition in a simplified hypothetical case in which father's occupation is the only circumstance, and the logarithm of income is the outcome of interest.[23] The objective is testing whether the parameters defining the income distribution are significantly different when the population is split into two subgroups.

Following the steps described in the main text, we set a confidence level ($\alpha = 0.01$), and in order to obtain a graphical intuition of the instability of the parameters, a lower order of the polynomial ($\omega = 3$), hence using four parameters to estimate the log income distribution. We generate a mock dataset to split incomes according to father's occupation, which takes six categories, ordered from smaller associated expected income to higher associated expected income.

In Figure A1, we show the values of the parameters in the Bernstein polynomial associated with each split. Beginning from the left-hand side in both plots, the first four points represent the parameters associated with the nodes created when we split the population into two groups: those whose father's occupation is 1 (right-hand plot) and the rest, that is, those whose father's occupation is 2–6 (left-hand plot). As we move to the right through the horizontal axis, we generate other splits, moving observations associated to categories in father's occupation from one node to another, changing the resulting conditioned distributions. It is evident from Figure A1 that when transitioning observations from one terminal node to another, parameters undergo a change in magnitude. However, it is not immediately apparent which partition exhibits the most statistically significant parameter instability, that is, which occupational category should be selected as the splitting point.

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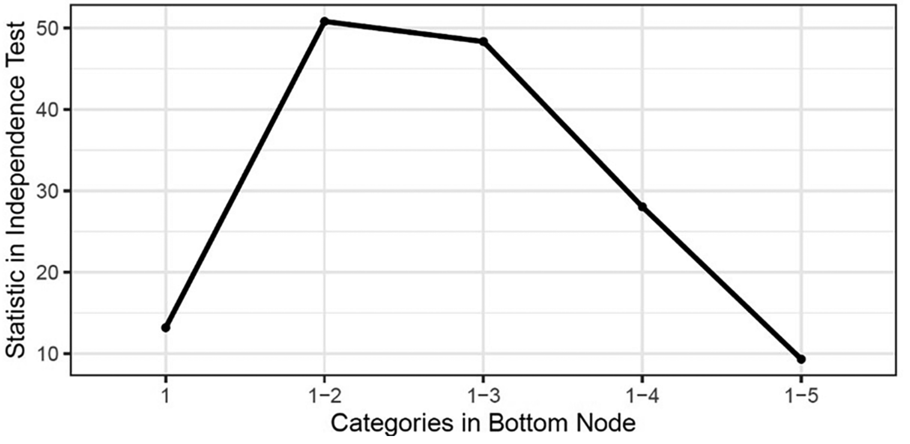

Figure A2 *M*-fluctuation quadratic test statistics. *Source*: Own elaboration on NIDS 5.

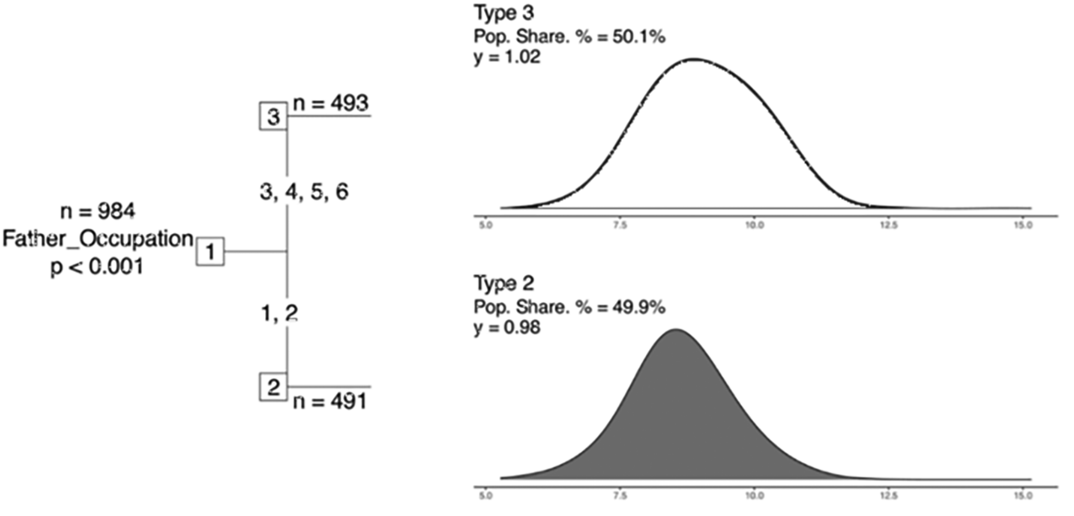

Figure A3 Transformation tree (example). *Source*: Own elaboration on NIDS 5.

That selection is guided by the *M*-fluctuation test. Figure A2 shows the value of the statistics for the tests described in step 4. The higher value (associated with a smaller *p*-value) is achieved when the bottom node has categories 1 and 2—that is, the splitting point, as confirmed in Figure A2. The population is thereby divided into two groups: those with father's occupation equal to 2 or less, and the rest, generating the simple tree in Figure A3.

This partition into two types allows us, for instance, to graphically explore Roemer's theory by plotting the CDFs of the outcome of interest by types (Figure A4). Here, the solid lines represent the ECDF, while the dashed lines represent the interpolation of the distribution predicted with the polynomial approximation.

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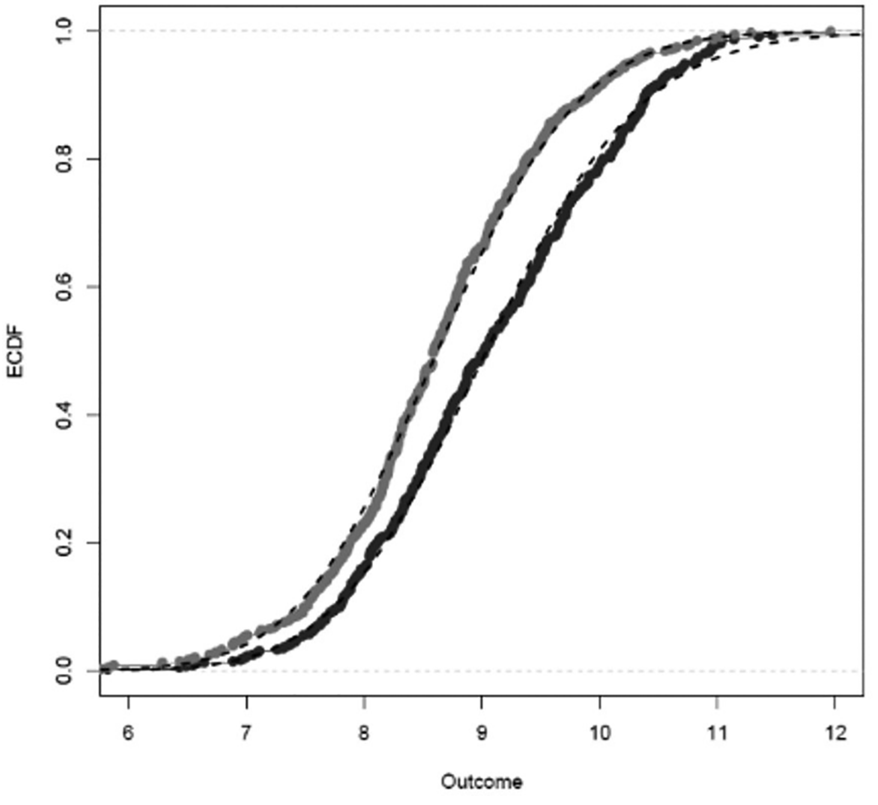

Figure A4 ECDFs (example). *Source*: Own elaboration.

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A.2 Additional tables and figures

Table A1 Descriptive income statistics for previous waves.

Country	Year	Mean	Gini	MLD	Top 1%	Top 10%	Bottom 40%
China	2010	4878	0.496	0.475	0.109	0.397	0.119
China	2012	6657	0.503	0.557	0.105	0.368	0.108
China	2014	6168	0.483	0.526	0.154	0.410	0.101
China	2016	8272	0.519	0.512	0.182	0.402	0.126
India	2005	5903	0.499	0.457	0.104	0.414	0.092
South Africa	2008	11,383	0.647	0.839	0.067	0.376	0.118
South Africa	2012	10,974	0.617	0.711	0.075	0.330	0.131
South Africa	2015	10,764	0.574	0.6	0.087	0.331	0.137
USA	1974	35,535	0.299	0.155	0.037	0.205	0.268
USA	1976	36,104	0.293	0.152	0.031	0.192	0.276
USA	1978	36,230	0.294	0.152	0.037	0.196	0.273
USA	1980	34,390	0.328	0.192	0.064	0.222	0.251
USA	1982	32,762	0.317	0.179	0.037	0.207	0.253
USA	1984	36,046	0.345	0.214	0.057	0.227	0.245
USA	1986	36,627	0.345	0.210	0.047	0.220	0.248
USA	1988	41,626	0.389	0.270	0.085	0.267	0.221
USA	1990	39,470	0.365	0.236	0.058	0.249	0.227
USA	1992	40,078	0.376	0.259	0.076	0.269	0.216
USA	1994	39,161	0.379	0.269	0.073	0.272	0.215
USA	1996	39,837	0.367	0.256	0.065	0.284	0.197
USA	1998	42,838	0.392	0.296	0.084	0.286	0.195
USA	2000	44,494	0.384	0.268	0.069	0.287	0.195
USA	2002	45,158	0.395	0.297	0.095	0.302	0.190
USA	2004	46,249	0.406	0.308	0.077	0.284	0.194
USA	2006	45,494	0.431	0.369	0.082	0.299	0.184
USA	2008	45,110	0.403	0.305	0.070	0.276	0.197
USA	2010	43,192	0.393	0.296	0.065	0.268	0.197
USA	2012	44,111	0.386	0.286	0.057	0.254	0.203
USA	2014	45,608	0.395	0.290	0.052	0.250	0.200
USA	2016	48,057	0.393	0.315	0.054	0.271	0.180

Notes: Income units are in 2017 US dollars at PPP exchange rates. The last three columns represent the shares of income received by the top 1%, the top 10%, and the bottom 40% in the income distribution.
Source: CFPS, IHDS, NIDS and PSID.

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Table A2 Inherited inequality estimates for previous waves.

Country	Year	Gini ($\hat{y}_T$)	IR_T (Gini)	MLD ($\hat{y}_T$)	IR_T (MLD)	Top 1%	Top 10%	Bottom 40%
China	2010	0.266	53.6%	0.133	28.0%	0.046	0.292	0.247
China	2012	0.218	43.3%	0.124	22.3%	0.049	0.237	0.233
China	2014	0.207	42.9%	0.104	19.8%	0.058	0.238	0.288
China	2016	0.293	56.5%	0.209	40.8%	0.051	0.287	0.246
India	2005	0.306	61.3%	0.178	39.0%	0.046	0.278	0.172
South Africa	2008	0.533	82.4%	0.534	63.6%	0.037	0.279	0.182
South Africa	2012	0.479	77.6%	0.405	57.0%	0.032	0.237	0.231
South Africa	2015	0.385	67.1%	0.248	41.3%	0.024	0.193	0.278
USA	1970	0.137	45.7%	0.032	20.1%	0.013	0.122	0.401
USA	1972	0.139	45.7%	0.034	20.5%	0.014	0.121	0.402
USA	1974	0.125	41.8%	0.027	17.4%	0.011	0.108	0.409
USA	1976	0.117	39.9%	0.024	15.8%	0.013	0.110	0.429
USA	1978	0.117	39.8%	0.024	15.8%	0.011	0.109	0.428
USA	1980	0.141	43.0%	0.035	18.2%	0.018	0.122	0.411
USA	1982	0.130	41.0%	0.030	16.8%	0.013	0.114	0.418
USA	1984	0.138	40.0%	0.034	15.9%	0.013	0.116	0.414
USA	1986	0.147	42.6%	0.037	17.6%	0.012	0.115	0.411
USA	1988	0.164	42.2%	0.048	17.8%	0.016	0.126	0.393
USA	1990	0.147	40.3%	0.037	15.7%	0.013	0.121	0.397
USA	1992	0.153	40.7%	0.043	16.6%	0.013	0.113	0.383
USA	1994	0.178	47.0%	0.081	30.1%	0.018	0.130	0.377
USA	1996	0.162	44.1%	0.051	19.9%	0.017	0.139	0.342
USA	1998	0.195	49.7%	0.086	29.1%	0.025	0.141	0.332
USA	2000	0.158	41.1%	0.046	17.2%	0.018	0.142	0.348
USA	2002	0.192	48.6%	0.079	26.6%	0.024	0.148	0.330
USA	2004	0.200	49.3%	0.090	29.2%	0.018	0.139	0.331
USA	2006	0.192	44.5%	0.108	29.3%	0.027	0.145	0.372
USA	2008	0.180	44.7%	0.076	24.9%	0.018	0.133	0.371
USA	2010	0.168	42.7%	0.057	19.3%	0.018	0.127	0.362
USA	2012	0.180	46.6%	0.113	39.5%	0.025	0.141	0.378
USA	2014	0.165	41.8%	0.054	18.6%	0.013	0.121	0.391
USA	2016	0.175	44.5%	0.080	25.4%	0.024	0.151	0.339

Notes: Inherited inequality estimated as $I(\hat{y})/I(y)$ (see Section 3). The last three columns represent the shares of $\hat{y}_T$ received by the top 1%, the top 10%, and the bottom 40% in the $\hat{y}_T$ distribution.
Source: CFPS, IHDS, NIDS and PSID.

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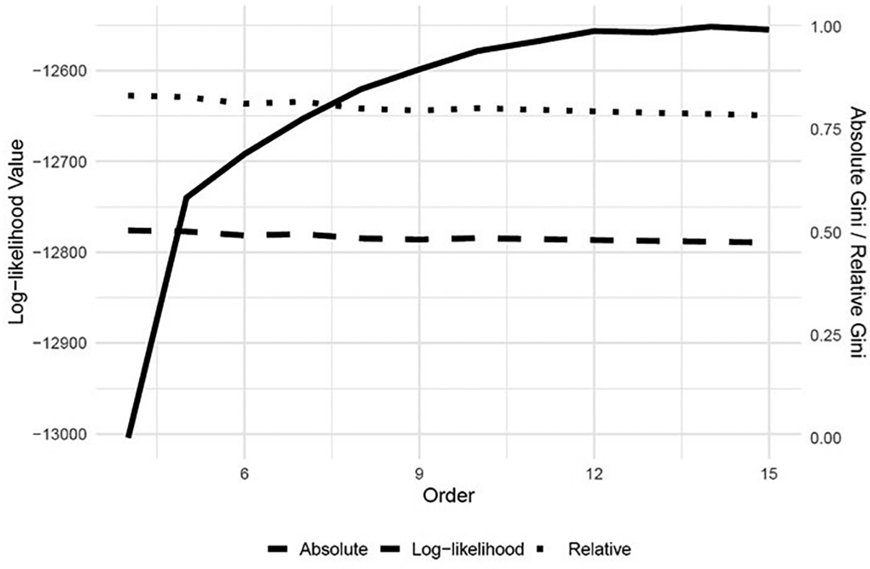

Figure A5 Sensitivity of inherited inequality to the Bernstein polynomial order in South Africa. *Notes*: The plot shows the log-likelihood value (solid line) associated with different Bernstein polynomial order values. We also show absolute (dotted line) and relative (dashed line) inherited inequality Gini values estimated with different Bernstein polynomial order values. The figure shows the average after 50 repetitions. *Source*: NIDS (2017).

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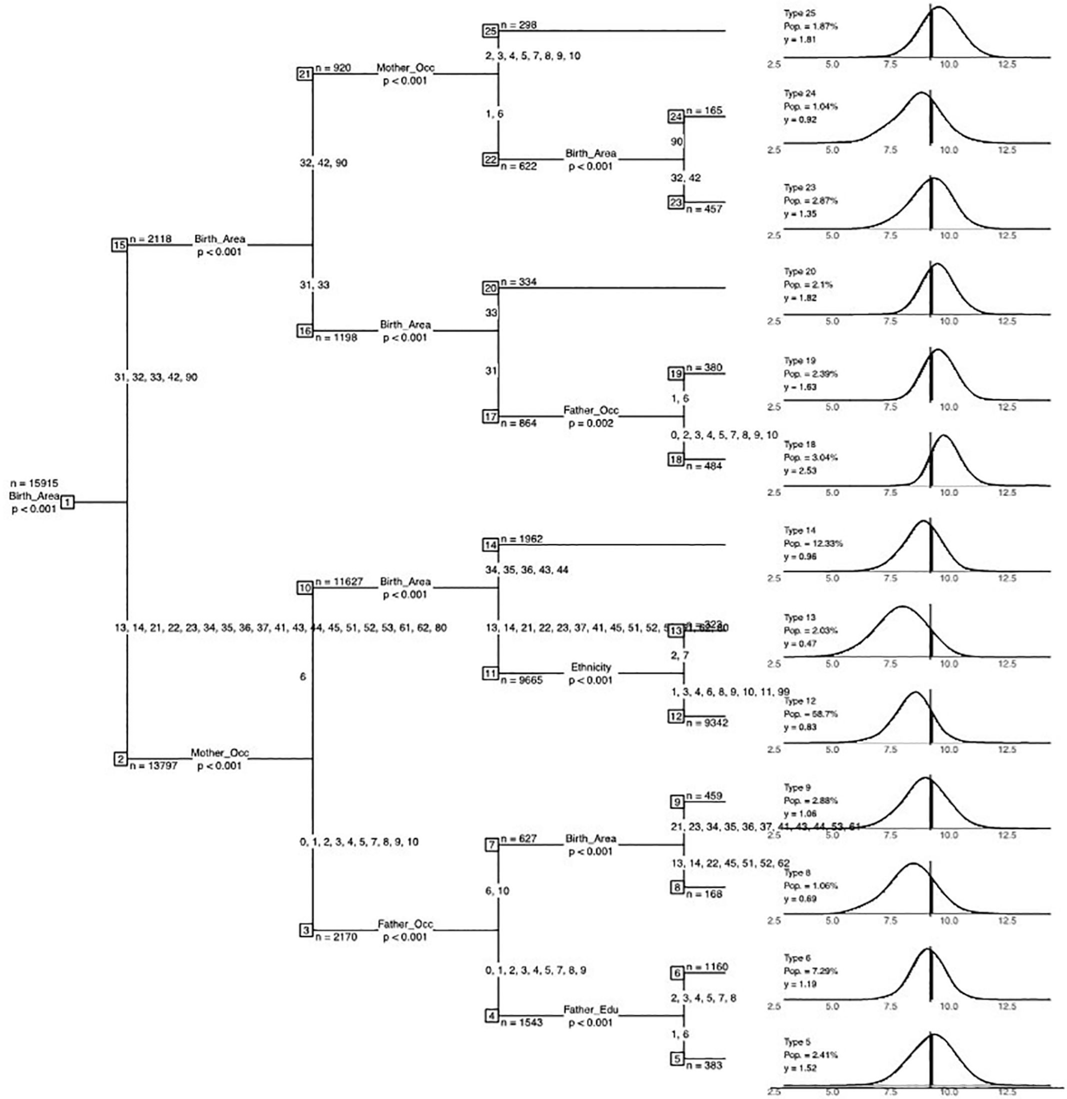

Figure A6 Transformation tree for China. *Notes*: Splitting nodes show their sample size and the *p*-value associated with the split. Circumstance categories are Gender (0 Female, 1 Male), Ethnicity (1 Han, 2 Mongol, 3 Hui, 4 Tibetan, 5 Miao, 7 Yi, 8 Zhuang, 9 Bouyei, 10 Korean, 11 Manchu, 99 Other), Birth area (13 Hebei, 14 Shanxi, 21 Liaoning, 22 Jilin, 23 Heilongjiang, 31 Shanghai, 32 Jiangsu, 33 Zhejiang, 34 Anhui, 35 Fujian, 36 Jiangxi, 37 Shandong, 41 Henan, 42 Hubei, 43 Hunan, 44 Guangdong, 45 Guangxi Zhuang Autonomous Region, 51 Sichuan, 52 Guizhou, 53 Yunnan, 61 Shaanxi, 62 Gansu, 80 Not available, 90 Other), Parents' education (1 Illiterate/semi-literate, 2 Primary school, 3 Junior high school, 4 Senior high school/secondary school/technical school/vocational senior school, 5 3-year college, 6 4-year college/bachelor's degree, 7 Master's degree, 8 Doctoral degree), Parents' occupations (0 Armed forces, 1 Managers, 2 Professionals, 3 Technicians and associate professionals, 4 Clerks, 5 Services and sales workers, 6 Agricultural, forestry and fishery workers, 7 Craft and trade workers, 8 Plant and machine operators and assemblers, 9 Elementary occupations, 10 Unemployed). The panels on the right display the log-density of type-specific incomes. The labels indicate the share of the population that each type represents (Pop.) and their average income relative to the overall sample mean ($y = 1$), which is also depicted as a vertical black line in the log-density plot. *Source*: CFPS (2018).

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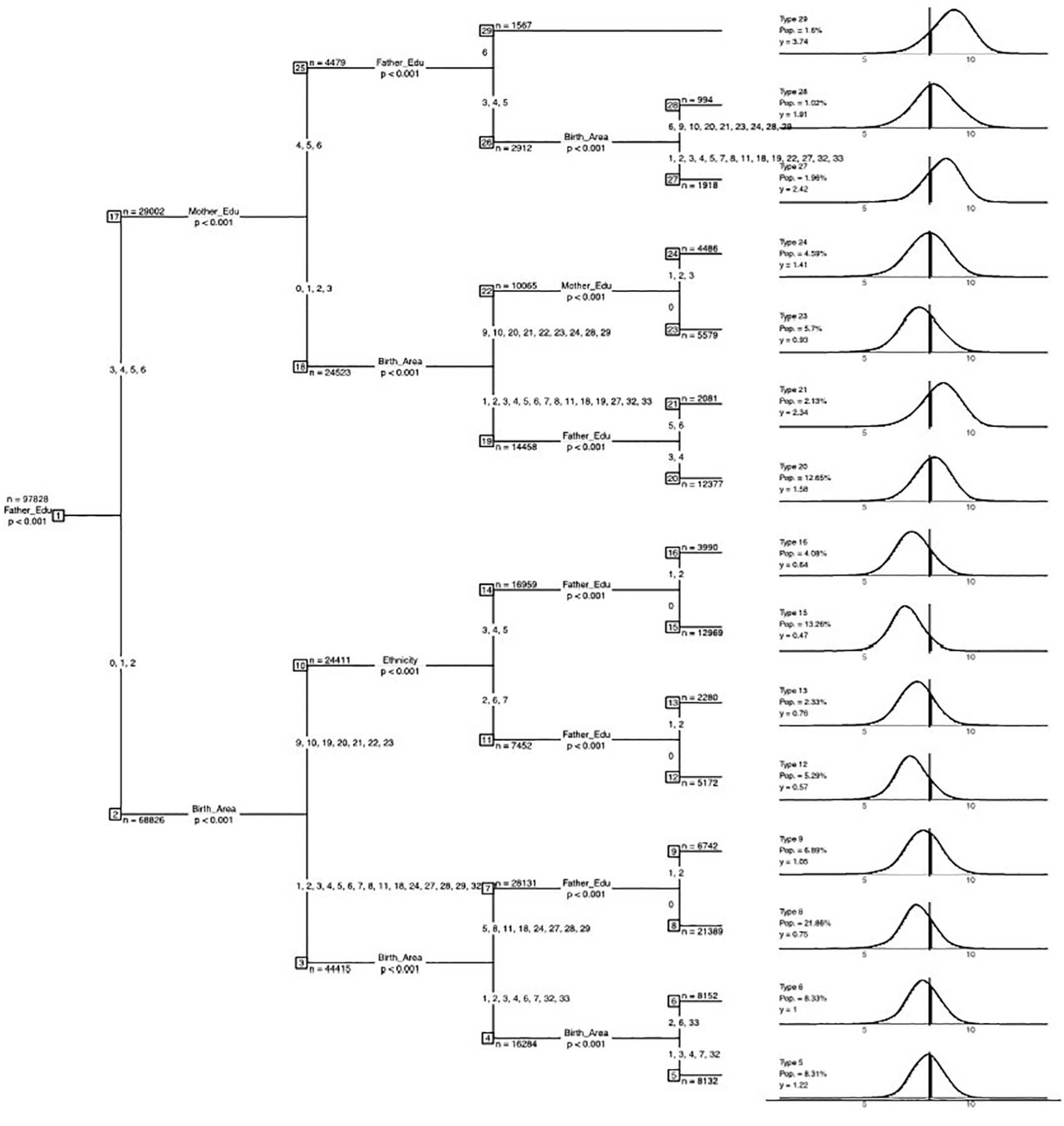

Figure A7 Transformation tree for India. *Notes*: Splitting nodes show their sample size and the p-value associated with the split. Circumstance categories are Gender (0 Female, 1 Male), Ethnicity (2 Forward caste, 3 Other Backward castes (OBC), 4 Dalit, 5 Adivasi, 6 Muslim, 7 Christian/Sikh/Jain), Parents' education (0 None, 1 Incomplete primary, 2 Complete primary, 3 Incomplete secondary, 4 Complete secondary, 5 Higher secondary, 6 Post-secondary or higher), Birth area (1 Jammu and Kashmir, 2 Himachal Pradesh, 3 Punjab, 4 Another state, 5 Uttarakhand, 6 Haryana, 7 Delhi, 8 Rajasthan, 9 Uttar Pradesh, 10 Bihar, 11 Overseas, 18 Northeast, 19 West Bengal, 20 Jharkhand, 21 Orissa, 22 Chhattisgarh, 23 Madhya Pradesh, 24 Gujarat, 27 Maharashtra, 28 Andhra Pradesh, 29 Karnataka, 32 Kerala, 33 Tamil Nadu). The panels on the right display the log-density of type-specific incomes. The labels indicate the share of the population that each type represents (Pop.) and their average income relative to the overall sample mean ($y = 1$), which is also depicted as a vertical black line in the log-density plot. *Source*: IHDS (2012).

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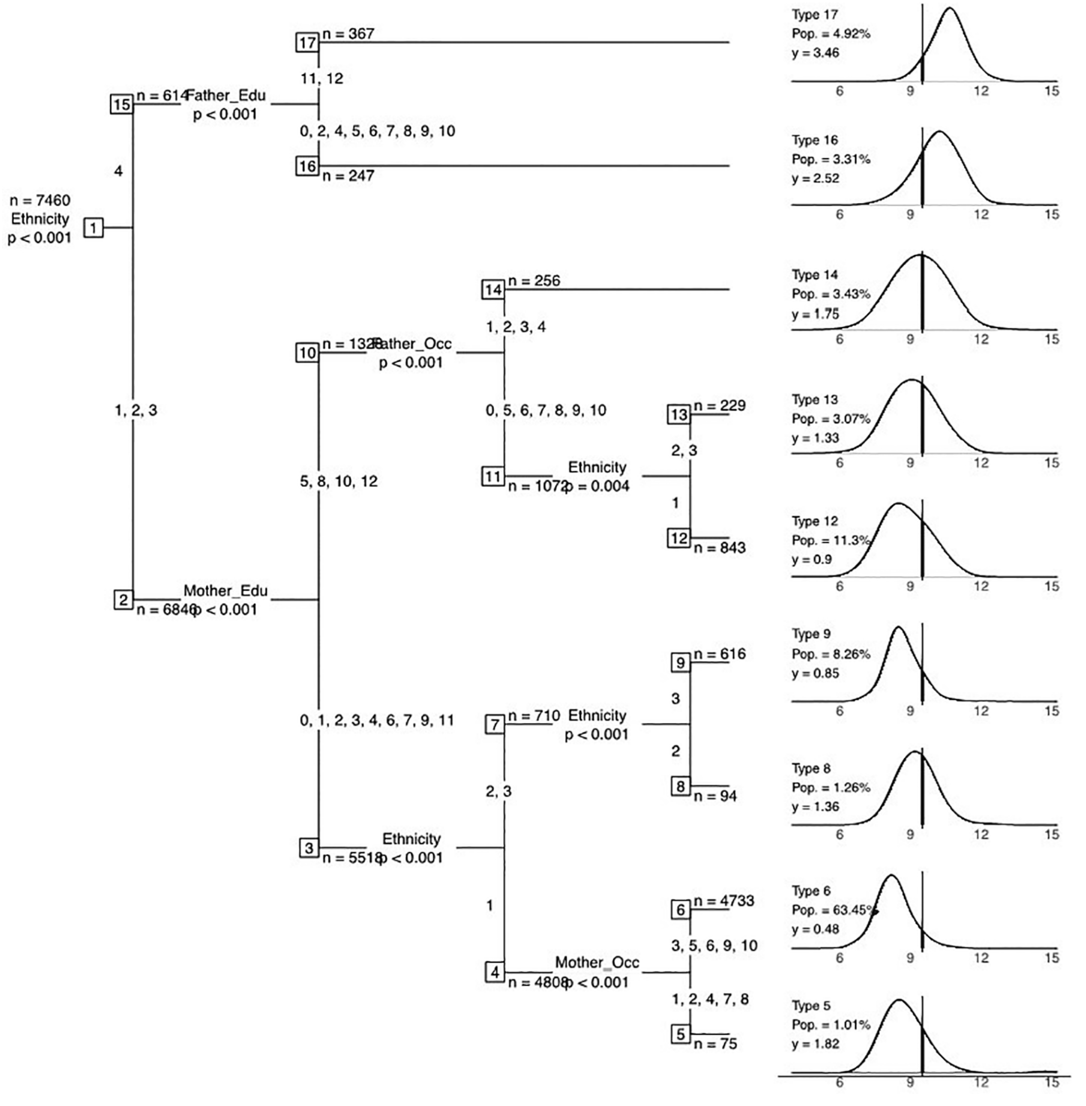

Figure A8 Transformation tree for South Africa. *Notes*: Splitting nodes show their sample size and the *p*-value associated with the split. Circumstance categories are Gender (0 Female, 1 Male), Ethnicity (1 African, 2 Asian/Indian, 3 Coloured, 4 White), Parents' education (0 Zero, 1 Grade 1, 2 Grade 2, 3 Grade 3, 4 Grade 4, 5 Grade 5, 6 Grade 6, 7 Grade 7, 8 Grade 8, 9 Grade 9, 10 Grade 10, 11 Grade 11, 12 Grade 12), Parents' occupations (0 Military, 1 Managers, 2 Professionals, 3 Technicians and associate professionals, 4 Clerical support, 5 Service and sales, 6 Farm, forest, fishery, 7 Craft, 8 Operators, 9 Elementary, 10 Others). The panels on the right display the log-density of type-specific incomes. The labels indicate the share of the population that each type represents (Pop.) and their average income relative to the overall sample mean ($y = 1$), which is also depicted as a vertical black line in the log-density plot. *Source*: NIDS (2017).

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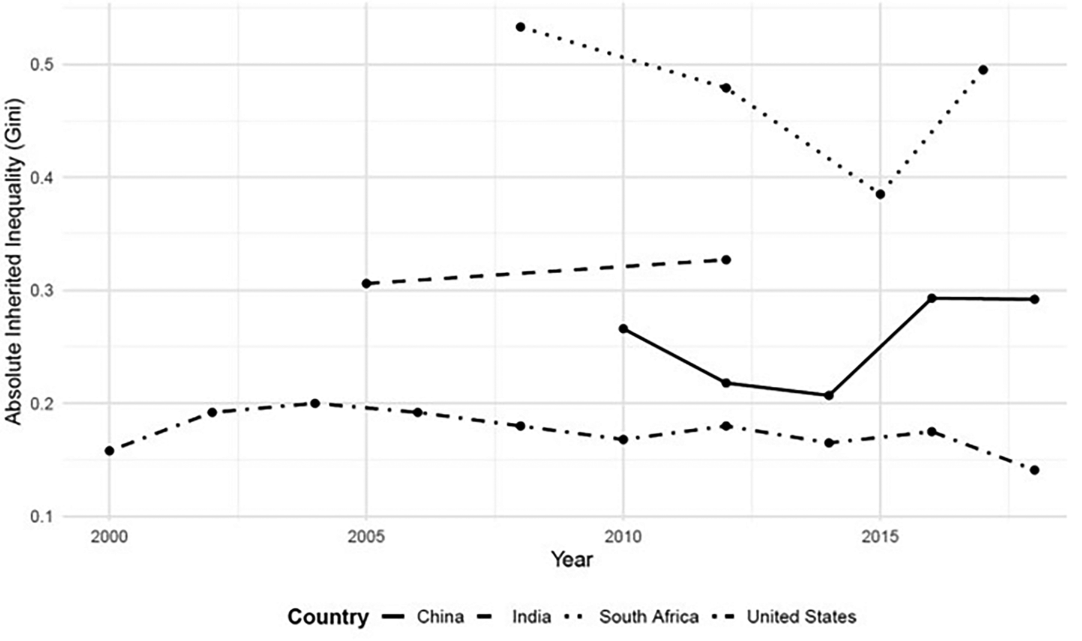

Figure A9 Time trends of inherited inequality since 2000. *Notes*: Inherited inequalities measured as the Gini index of the inherited component of income. *Source*: CFPS, IHDS, PSID and NIDS.

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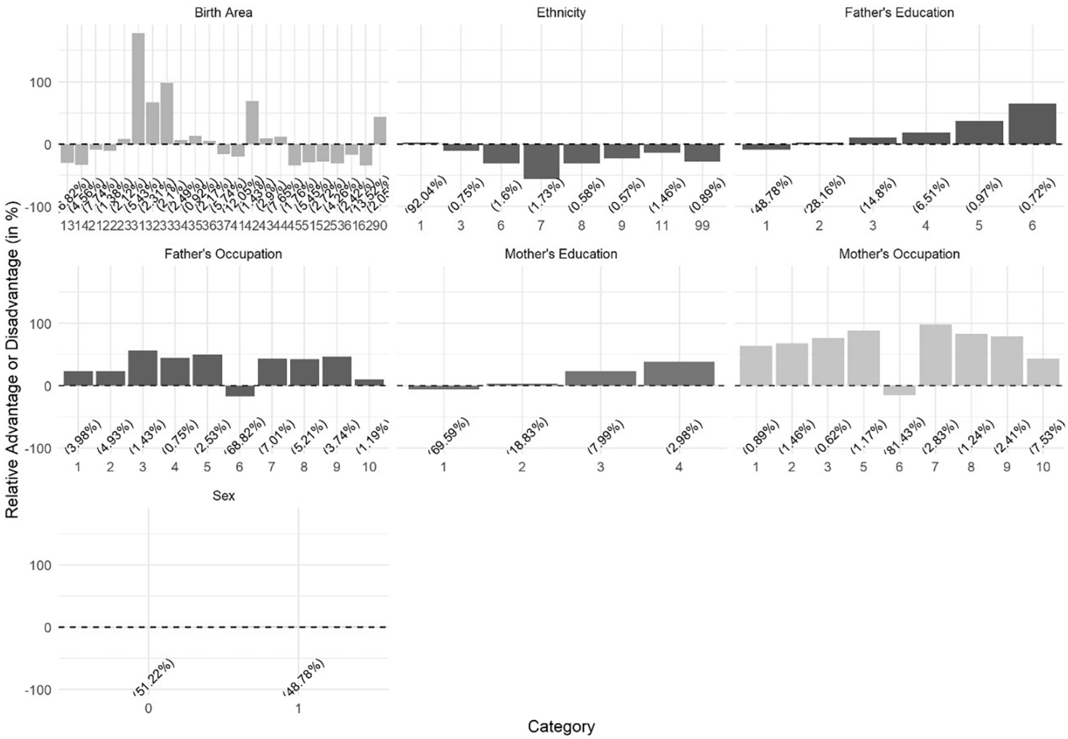

Figure A10 Marginal effects of circumstances on the inherited component of income: China (2018). *Notes*: Values on the vertical axis represent the marginal effects associated with each category, computed as described in the main text. Circumstance categories are Gender (0 Female, 1 Male), Ethnicity (1 Han, 2 Mongol, 3 Hui, 4 Tibetan, 5 Miao, 7 Yi, 8 Zhuang, 9 Bouyei, 10 Korean, 11 Manchu, 99 Other), Birth area (13 Hebei, 14 Shanxi, 21 Liaoning, 22 Jilin, 23 Heilongjiang, 31 Shanghai, 32 Jiangsu, 33 Zhejiang, 34 Anhui, 35 Fujian, 36 Jiangxi, 37 Shandong, 41 Henan, 42 Hubei, 43 Hunan, 44 Guangdong, 45 Guangxi Zhuang Autonomous Region, 51 Sichuan, 52 Guizhou, 53 Yunnan, 61 Shaanxi, 62 Gansu, 80 Not available, 90 Other), Parents' education (1 Illiterate/semi-literate, 2 Primary school, 3 Junior high school, 4 Senior high school/secondary school/technical school/vocational senior school, 5 3-year college, 6 4-year college/bachelor's degree, 7 Master's degree, 8 Doctoral degree), Parents' occupations (0 Armed forces, 1 Managers, 2 Professionals, 3 Technicians and associate professionals, 4 Clerks, 5 Services and sales workers, 6 Agricultural, forestry and fishery workers, 7 Craft and trade workers, 8 Plant and machine operators and assemblers, 9 Elementary occupations, 10 Unemployed). Numbers in parentheses denote population shares in each category. We are not showing categories populated by less than 0.5% of the sample size. *Source*: CFPS (2018).

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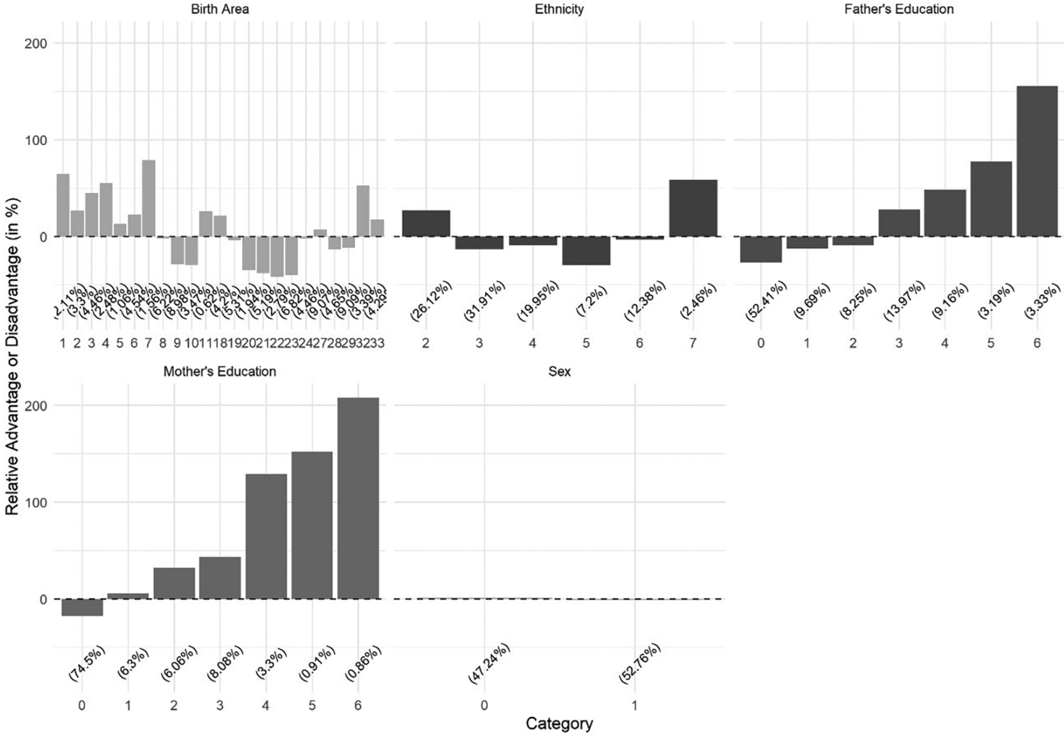

Figure A11 Marginal effects of circumstances on the inherited component of income: India (2012). *Notes*: Values on the vertical axis represent the marginal effects associated with each category, computed as described in the main text. Circumstance categories are Gender (0 Female, 1 Male), Ethnicity (2 Forward caste, 3 Other Backward castes (OBC), 4 Dalit, 5 Adivasi, 6 Muslim, 7 Christian/Sikh/Jain), Parents' education (0 None, 1 Incomplete primary, 2 Complete primary, 3 Incomplete secondary, 4 Complete secondary, 5 Higher secondary, 6 Post-secondary or higher), Birth area (1 Jammu and Kashmir, 2 Himachal Pradesh, 3 Punjab, 4 Another state, 5 Uttarakhand, 6 Haryana, 7 Delhi, 8 Rajasthan, 9 Uttar Pradesh, 10 Bihar, 11 Overseas, 18 Northeast, 19 West Bengal, 20 Jharkhand, 21 Orissa, 22 Chhattisgarh, 23 Madhya Pradesh, 24 Gujarat, 27 Maharashtra, 28 Andhra Pradesh, 29 Karnataka, 32 Kerala, 33 Tamil Nadu). Numbers in parentheses denote population shares in each category. We are not showing categories populated by less than 0.5% of the sample size. *Source*: IHDS (2012).

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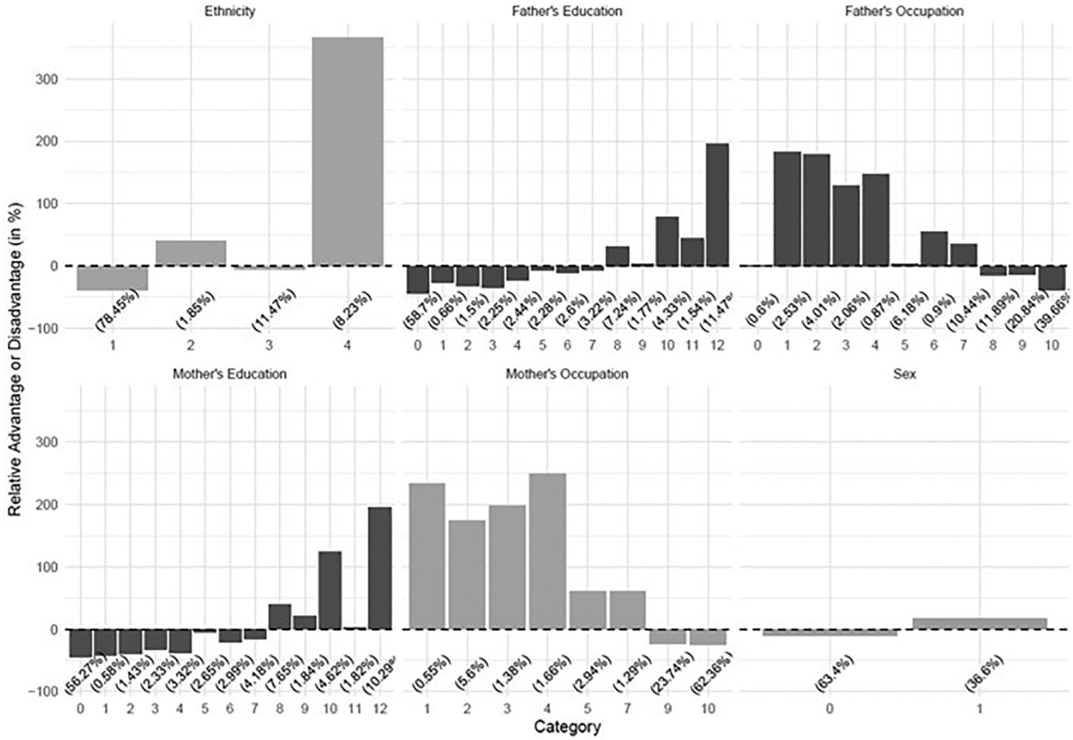

Figure A12 Marginal effects of circumstances on the inherited component of income in South Africa (2017).
Notes: Values on the vertical axis represent the marginal effects associated with each category, computed as described in the main text. Circumstance categories are Gender (0 Female, 1 Male), Ethnicity (1 African, 2 Asian/Indian, 3 Coloured, 4 White), Parents' education (0 Zero, 1 Grade 1, 2 Grade 2, 3 Grade 3, 4 Grade 4, 5 Grade 5, 6 Grade 6, 7 Grade 7, 8 Grade 8, 9 Grade 9, 10 Grade 10, 11 Grade 11, 12 Grade 12), Parents' occupations (0 Military, 1 Managers, 2 Professionals, 3 Technicians and associate professionals, 4 Clerical support, 5 Service and sales, 6 Farm, forest, fishery, 7 Craft, 8 Operators, 9 Elementary, 10 Others). Numbers in parentheses denote population shares in each category. We are not showing categories populated by less than 0.5% of the sample size. *Source*: NIDS (2017).

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